Unit -1

Introduction to Software Engineering

Nature of a software [pg no.3 book R.P,7th edition]

- Software takes dual role. It is a **product** and at the same time ,the **vehicle for delivering a product.**
- Software is an information transformer, producing, managing , acquiring , modifying , displaying or transmitting into that Can be as simple as single bit or complex as a multimedia presentation derived from data acquired from dozens of independents sources.

Defining software

- **Software is-**

1. **Instruction** that when executed provide desired features, functions and performance.
2. **Data structure** that enable the program to adequately manipulate information.
3. **Descriptive information** in both hard copy and virtual forms that describes the operation and use of the program.

- Software is uses **logical** rather than physical system elements.

1.Software is developed not manufactured.

2.Software does not ‘wear out’

Fig.1 failure curve for hardware

- It indicates that hardware exhibits relatively high failure rate early in its life.
- Defects are corrected and failures are drops to a steady state level for some period of time.
- Software does not wear out but deteriorate(corrupt/spoil).

Software

- Software is nothing but a collection of computer programs and related documents that are intended to provide desired features, functionalities and better performance.
- Software products may be-

I. **Generic :** that means developed to be sold to a range of different customers.

Eg.PC s/w such as Excel or word.

II. Custom: that means developed for a single customer according to their specification.

Eg. Air traffic monitoring

Software Engineering

- **s/w engineering** is a discipline in which theories, methods and tools are applied to develop professional s/w product .
- In software engineering the **systematic and organized** approach is adapted.
- The definition of s/w engineering is based on two terms-

1. **Discipline:** for finding the solution to the problem an engg. applies appropriate theories, methods and tools.

2.Product: the s/w products get developed after following systematic theories, methods and tools along with the appropriate management activities.

Nature of software: [pg no 3 book R.P]

- Software can be applied in a situation for which a predefined set of procedural steps (algorithms) exist.

- Based on a complex growth of s/w it can be classified into following-

1. **System software:** It is collection of programs written to service other program.
2. **Application software:** It consist of standalone program that are developed for specific business head.
3. **Embedded software:** this category consist of program that can reside within a product or system.
4. **Web Applications:** this s/w consist of various web pages that can be retrieved by a browser.

5.Engineering/scientific s/w: this category of s/w has wide range of programs from astronomy to volcano logy ,from automotive stress analysis to space shuttle orbital dynamics and from molecular energy to automated manufacturing.

This s/w is based on complex numeric computation.

6.Artificial Intelligence s/w: this type of s/w are based on knowledge based expert system.

Typically this s/w is useful in robotics, expert system, image and voice recognition, artificial neural n/w, and game playing

Software Engineering Practice

- Reason for Existence
- Keep it Simple
- Vision
- The Produce is getting consumed
- Be open to future
- Plan to Reuse
- Think

Layered Technology [pg no-14]

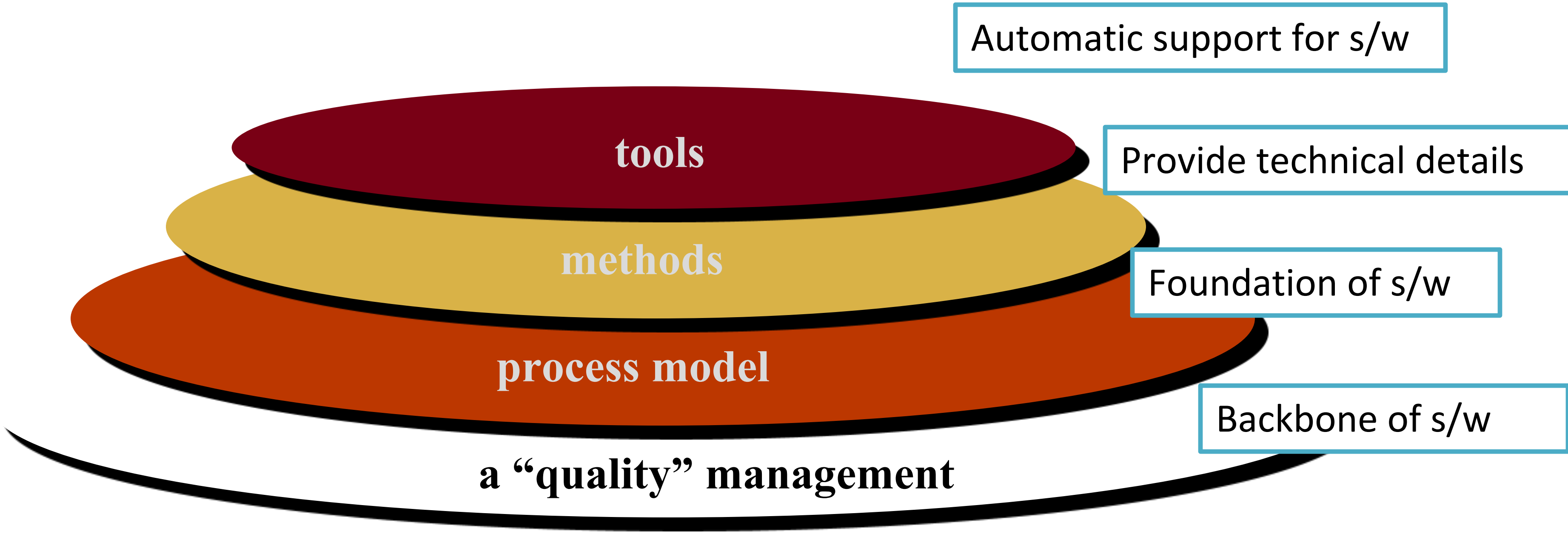


Figure 2. software engineering layers

- It is a layered technology.
- Any s/w can be developed using these layered approaches.
- So software engg is a combination of process, method and tools for development of quality s/w.

Questions

- Define Software Engineering. what are the software characteristics? What are the various categories of software?

- Define term software & software engineering. Software does not wear out state whether this statement is true or false. Justify your answer.

MCQ

- 1. ______ software is a collection of programs written to service other.

  a. Application

  b. System

  c. Engineering

  d. Embedded

MCQ

- 2. Software is a Process and ____.

a. Program

b. Design

c. Product

d. None of the above

MCQ

- Robotics, expert system and pattern recognition are the applications of ____.

a. system software

b. Web programming

c. AI

d. Game programming

The Software process

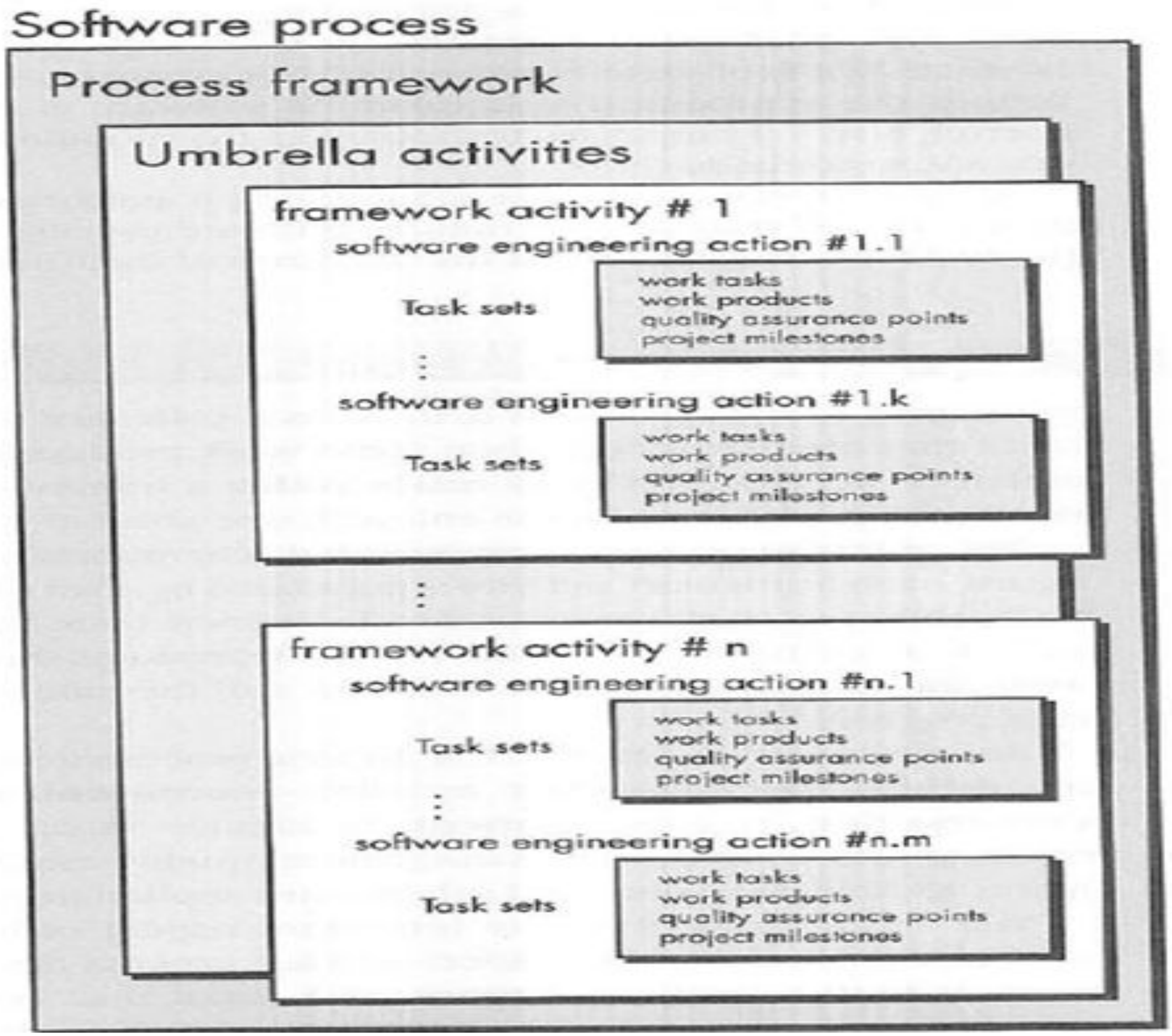


Figure 3 software process

- Software process can be defined as the structured set of activities.
- Common process framework is established by dividing overall framework activities into small number of framework activities.
- These small number of activities are applicable to the entire project irrespective of its size and complexity.
- A framework is a collection of task sets and these task sets consist of -

1. Collection of small work tasks.
2. Project milestones, deliverables.
3. s/w quality assurance point

- There are various activities called umbrella activities and these activities are associated throughout the development process.
- **Umbrella Activities :**

► Software project management

► Formal technical reviews

► Software quality assurance

► Software configuration management

► Work product preparation and production

► Reusability management

► Measurement

► Risk management

The umbrella activities include-

I. Software project tracking & control: Developing team has to access project plan and compared with predefined schedule.

II. Risk management: risk management is an event that may or may not occur. assesses risks that may affect outcome of project.

III. Software quality assurance: It is nothing but planned and systematic pattern of activities those are required to give guarantee of s/w quality.

IV. Formal technical review: IT is a meeting conducted by technical staff.

- The purpose of meeting is to detect quality problems and suggest quality improvement.

V. Measurement: It includes the effort required to measure the software (cost,line of code ,size).

vI. Software configuration management: It manages effect of change through out the s/w process.

VII. Reusability management: It defines criteria for product reuse.

VIII. Work product preparation and production:

It includes activities required to create documents, logs. Forms, lists and user manual for development software.

Framework Activities

- Communication
- Planning
- Modeling
  - Analysis of requirements
  - Design
- Construction
  - Code generation
  - Testing
- Deployment

Software Vs. Hardware

- Software is developed or engineered, it is not manufactured in the classical sense.

- Software doesn’t wear out.

- Although the industry is moving towards component based construction, most software continues to be custom built.

Questions

- Elaborate how software engineering is a layered technology?
- What are the various umbrella activities applied throughout a software project?
- What is software process framework? Explain in detail.

MCQ

- 1. software engineering, is a layered technology. It consist of ____ layers.
- A. two
- B. Three
- C. Four

2._______process framework activity is responsible for feedback.

a. Communication b.Modeling

c. Construction d. deployment

Software Myths [pg.no-21 to 24 book R.P.7th edition]

- Dictionary meaning of myth is fable stories.
- It is fiction, imagination or it is a thing or story whose existence is not verifiable.
- In relation with computer software myth is nothing but misinformation, misunderstanding or confusions propagate in software development field.
- In software development the myths can be considered as three level myths.

1. Management level myths

2. Customer level myths

3. Practitioner/developer level myths

Management level myths

- Myth 1:we already have a book that's full of standards and procedures for building software, wont that provide my people with everything they need to know?
- Myth 2: My people have state of the art software development tools, after all we buy them the newest computers.
- Myth 3: If we get behind schedule, we can add more programmers and catch up.
- Myth 4: if I decide to outsource the software project to a third party, I can just relax and let that firm build it.

Customer level myths

- Myth 1: A general statement of objectives is sufficient to begin writing programs we can fill in the details later.

- Myth 2: Project requirement continually changes, but change can be easily accommodated because software is flexible.

Practitioner/developer level myths

- Myth 1: once we write the program and get it to work, our job is done.
- Myth 2: Until I get the program “running” I have no way of assessing its quality.
- Myth 3: The only deliverable work product for a successful project is the working program.
- Myth 4: There is no need of documenting the software project, it unnecessarily slows down the development process.

Software crisis

- Most of the s/w engg refers the problem associate with s/w development as **"software crisis".**
- The cause of s/w crisis are linked to all the problems and complexities associated with development process.
- Following are some most encountered problems associated with development process-

1. Running out of time i.e. deadline crosses.
2. Over budget
3. s/w inefficient
4. Low quality
5. Not upto expectations of customers
6. No enough development team.

Need of process model

Each team member will understand what is the next activity and how to do it.
Thus process model will bring the definiteness and discipline in overall development process.

A generic process model

- Software process is collection of various activities.
- There are 5 generic process framework activities.

1. **Communication:** The software development process starts with communication between customer and developer.
2. **Planning:** It includes complete estimation and scheduling and tracking.

3.Modeling:it includes detail requirement analysis and project design (flowchart, algorithm)**.**

4.Construction: it includes coding and testing steps.

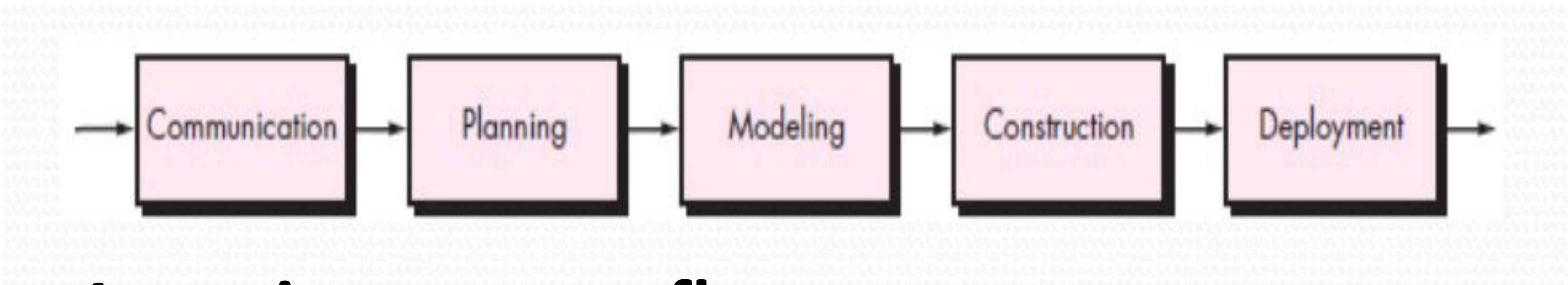
5.Deployment: it includes software delivery, support and feedback from customer.

Legacy software

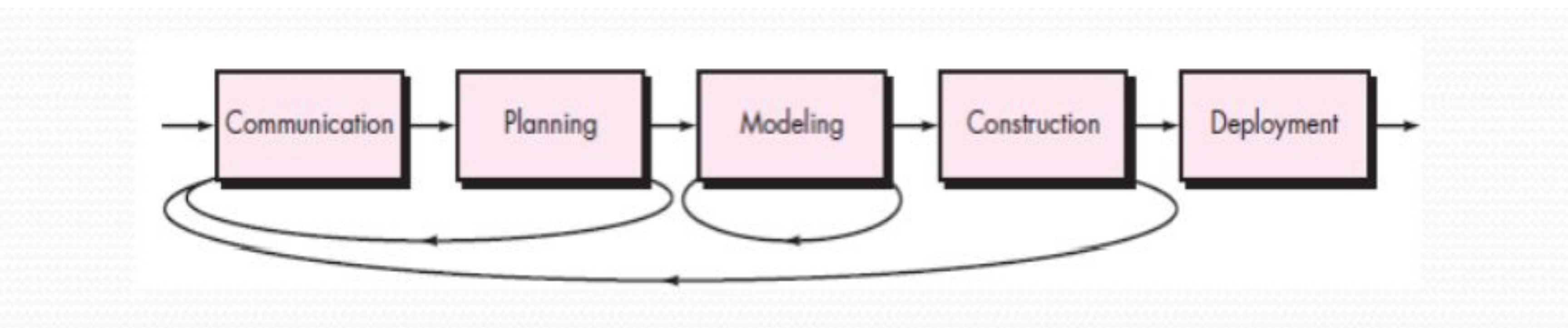
- The term legacy software refers to older s/w development that were poorly designed and documented.
- It had supported for many years.
- The legacy system may or may not remain in use.
- Even if it is no longer used, it may continue to impact the organization due to its historical role

Process Flow

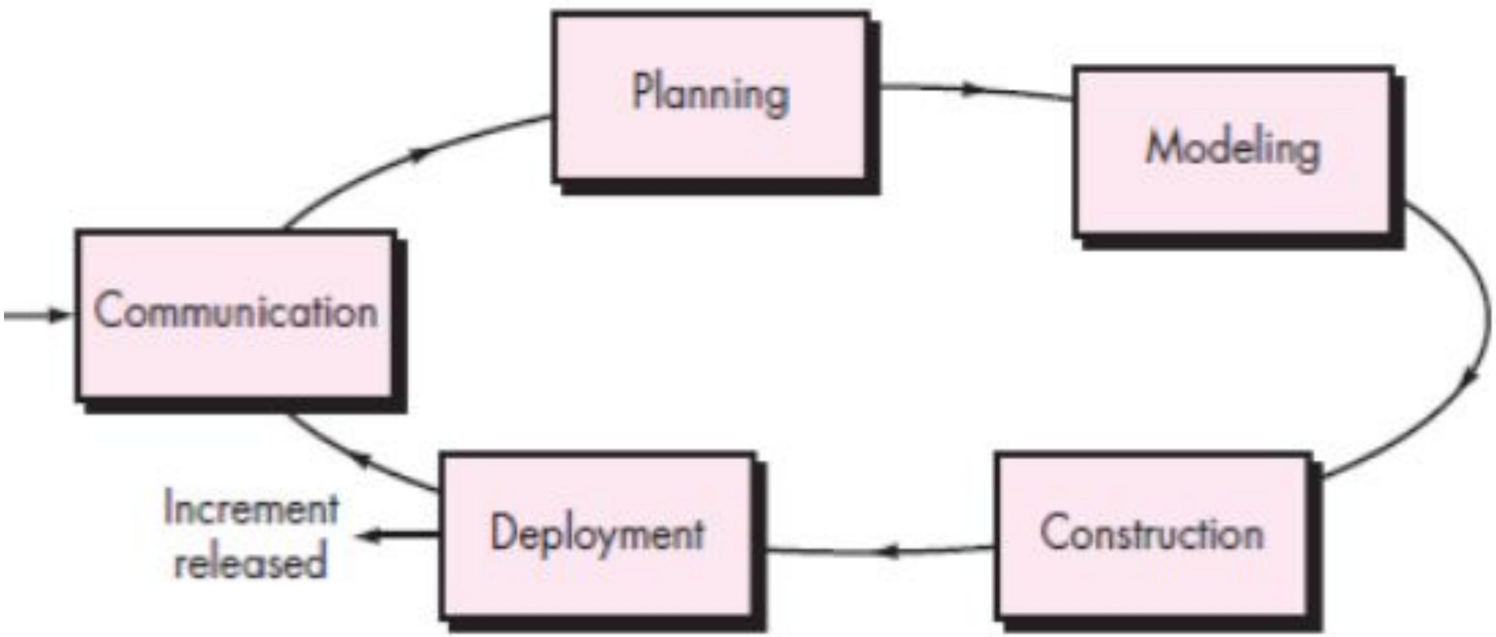
- **Linear process flow:** Execute flow from communication to deployment in sequence.



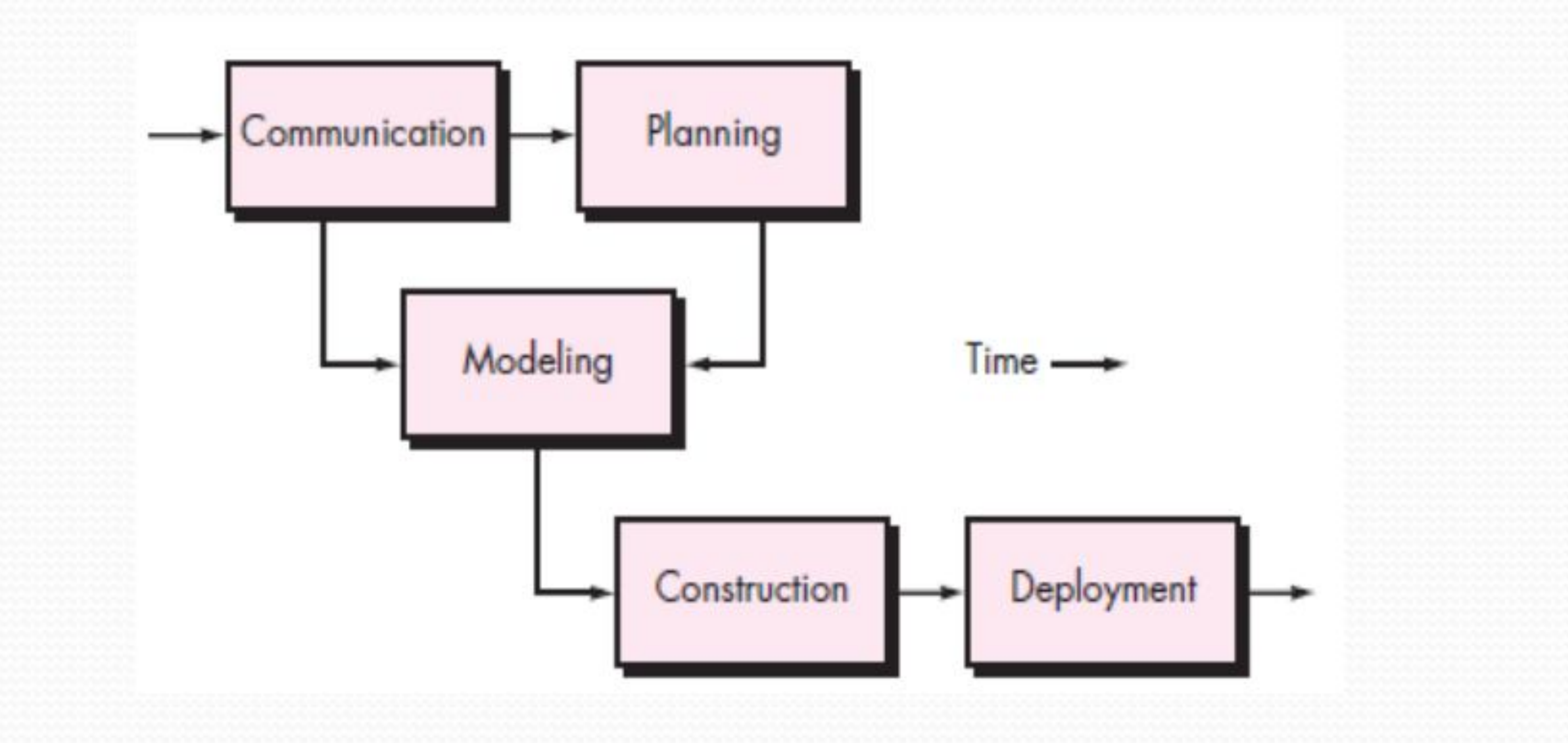
- **Iterative process flow:**



- **Evolutionary process flow:** executes the activities in a circular manner to provide a software.



- **Parallel process flow:** executes one or more activities in parallel with other activities.



Types of process models

- Generic process model
- Prescriptive process model
- Evolutionary process model

Generic process model

- A generic process model has-

1.Defining a framework activity:

 Work tasks for a simple software projects are-

- Make contacts with stakeholder via telephone.
- Discuss requirements and take notes.
- Organize notes in brief written statements of requirements.
- E-mail to stakeholder for review and approval.

 If project is complex, then go for :inception, elaboration, negotiation, specification and validation.

2.Identifying a task set:

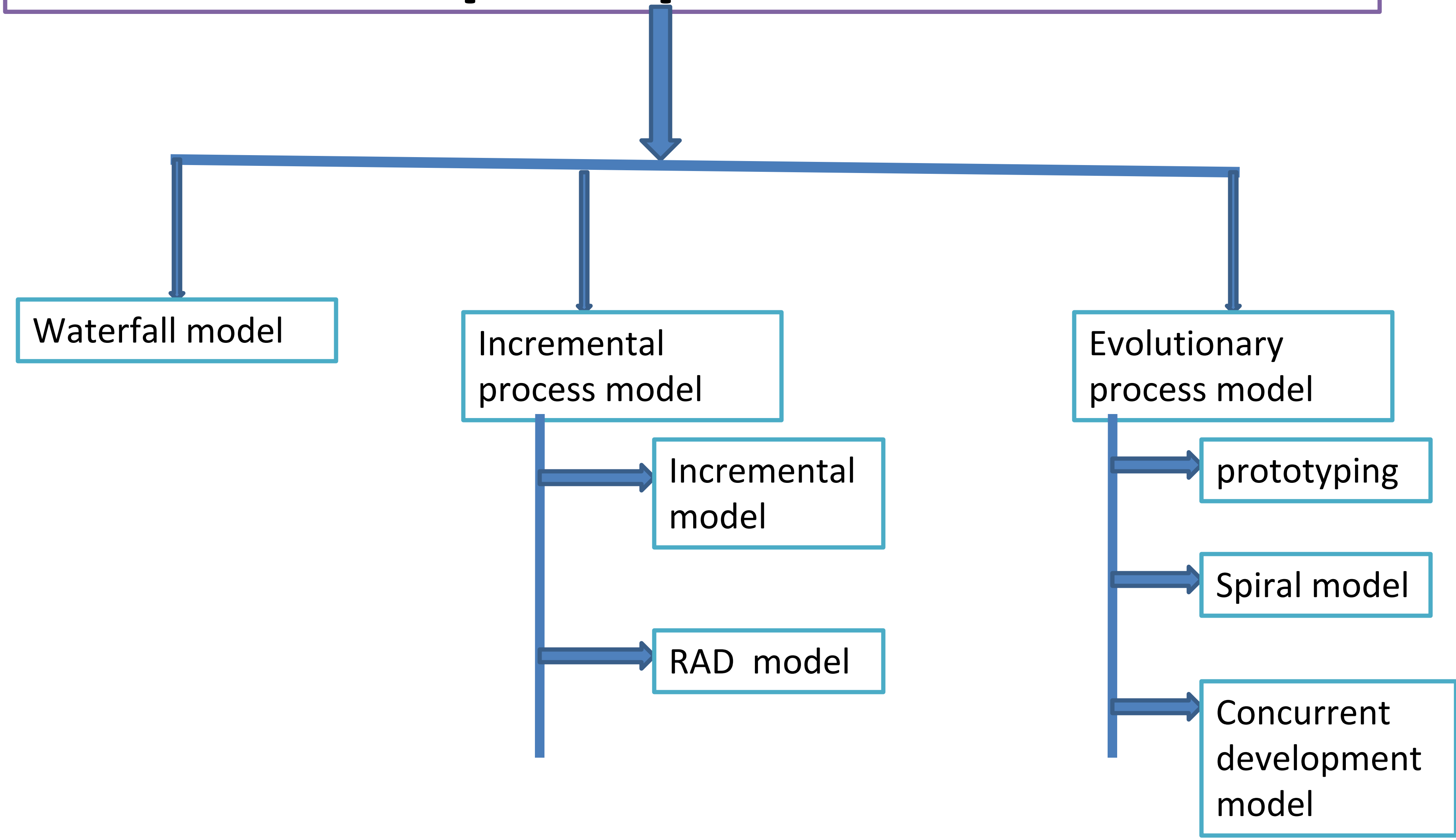
- Task sets are actual list of work to be done to accomplish the objective of s/w engineering action.
- Select suitable task set based on problem and project.

3.Process patterns: proven solutions for the similar problem

Prescriptive process model(SDLC)

- This process model define prescribed set of process elements and a predictable process workflow.
- Organize framework activities in a certain order
- Process framework activity with set of software engineering actions.
- Each actions in terms of a task set that identifies the work to be accomplished to meet the goals.
- The resultant process model should be adapted to accommodate the nature of the specific project , people doing the work , and the work environment.

Prescriptive process model



Waterfall model [pg no.39-40 book R.P]

- Waterfall model is also called as **"linear sequential model"** or **"classic life cycle model".**
- The software development will start with requirement gathering phase.
- then progresses through analysis, design, coding, testing and maintenance.

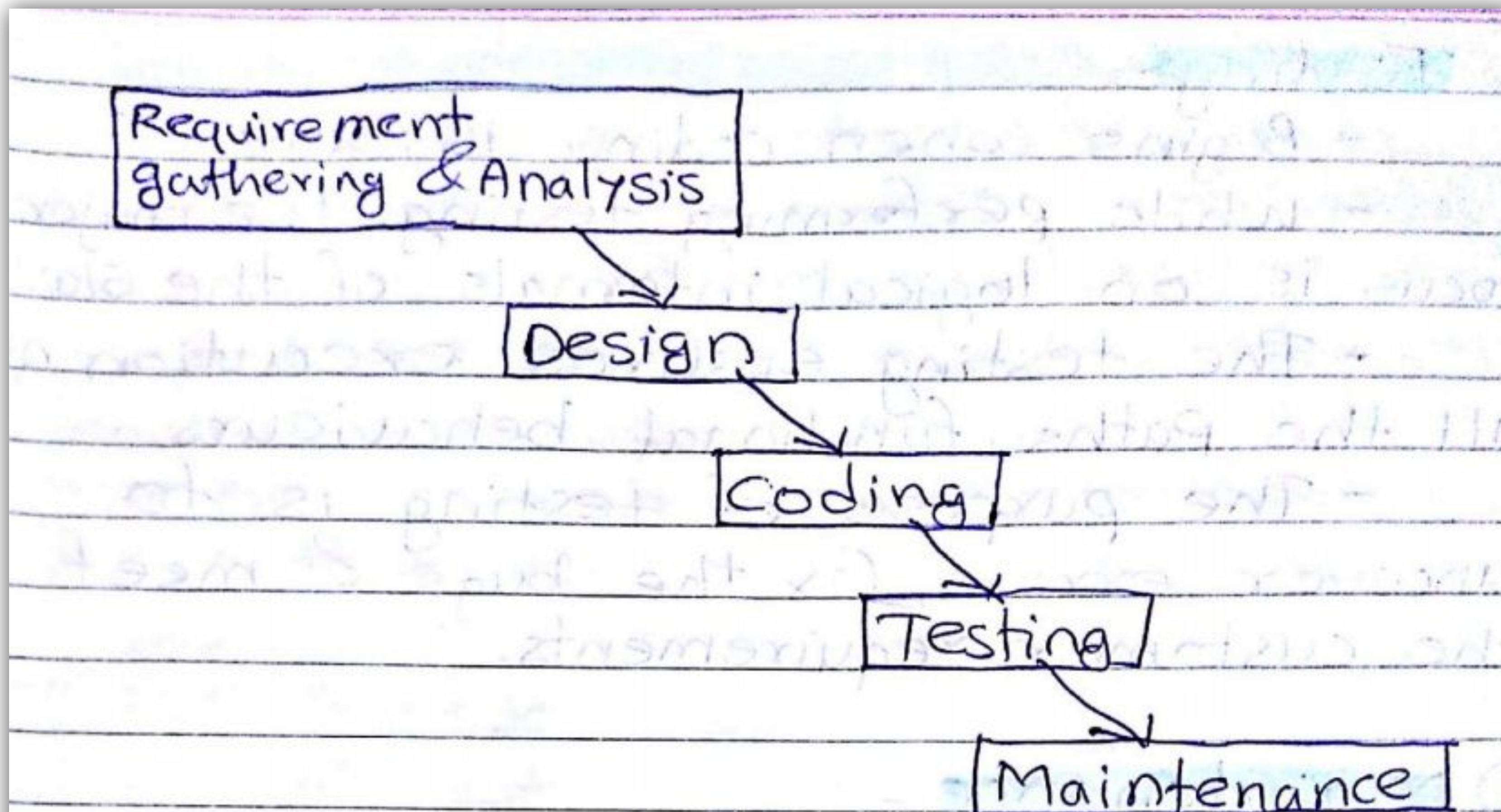
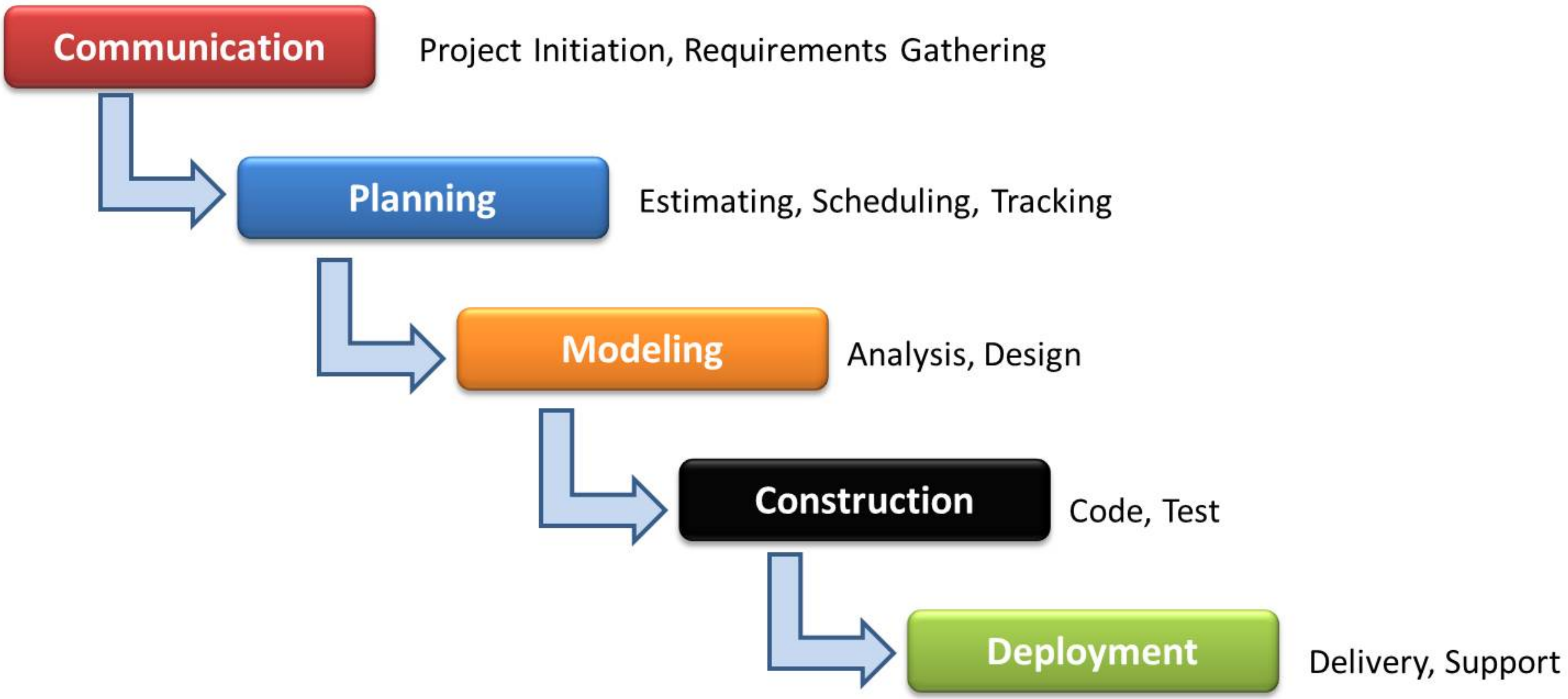


Figure 5: Waterfall Model



The Waterfall Model: A Traditional Approach of SDLC

1. **Requirement gathering and analysis:**

- In this phase basic requirements of the system must be understood by software engineer who is also called as **analyst.**
- The information domain function behavioral requirements of the system are understood.
- All the requirements are then well documented and discussed with customer.

2. Design:

Focuses on program attributes such as-

- Data structure
- Software architecture
- Interface representation
- Algorithmic details

3. Coding:

In this phase actual coding will start according to design.

4. Testing:

- Begins when coding will done.
- While performing testing more focus is on logical internals of the software.
- Testing ensures execution of all the paths, functional behaviors
- The purpose of testing is to uncover errors, fix the bugs and meet the customer requirements.

5. Maintenance:

- Longest life cycle phase.
- When the system is installed and put in practical use then error may get introduced.
- Correcting such errors and putting it in is the major purpose of maintenance activity.

Benefits:

- The waterfall model is **simple to implement**.
- For implementation of **small systems** waterfall model is useful.

Drawbacks:

- Difficult to follow sequential flow.
- Requirement analysis is done initially and sometimes it is not possible to state all the requirements explicitly in the beginning .
- The customer can see working model only at the end.
- Linear nature of waterfall model induced blocking state because certain tasks may be dependent on some previous task

Questions

- What are the elements of waterfall model? State its merits and demerits.
- What are customer myths? Discuss the reality of these myths?
- List and explain practitioner’s myth and its realities?

- 1. SDLC stands for _____.

- A. Software Document Listing collection
- B. Software Development Life cycle
- C. Software Deployment Life cycle
- D. System Development Life cycle

MCQ

- 2. Purpose of process is to deliver software _____.
- A. in time
- B. with acceptable quality
- C. that is cost efficient
- D. Both in time & with acceptable quality

Incremental Process model(eg RAD model)

- In this model the initial model with limited functionality is created for users understanding about s/w product and then this model is refined and expanded in later releases.
- **Phases in incremental model:**

1. Analysis
2. Design
3. Code
4. Test

- Incremental model delivers series of releases to the customer, these releases are called **increments**.
- The first increment is called **core product**. In this release basic requirements are implemented and then, in subsequent increments new requirements are added.

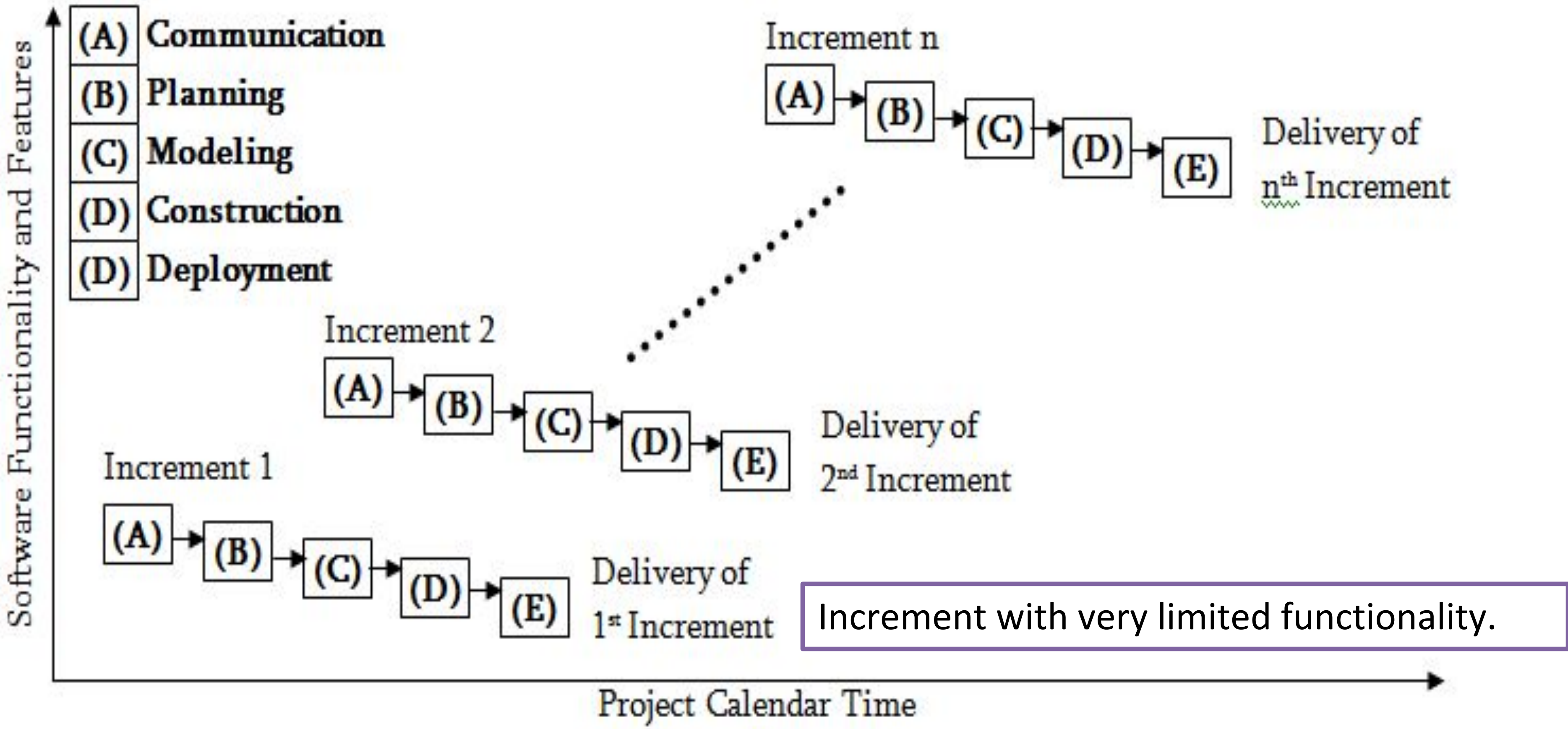
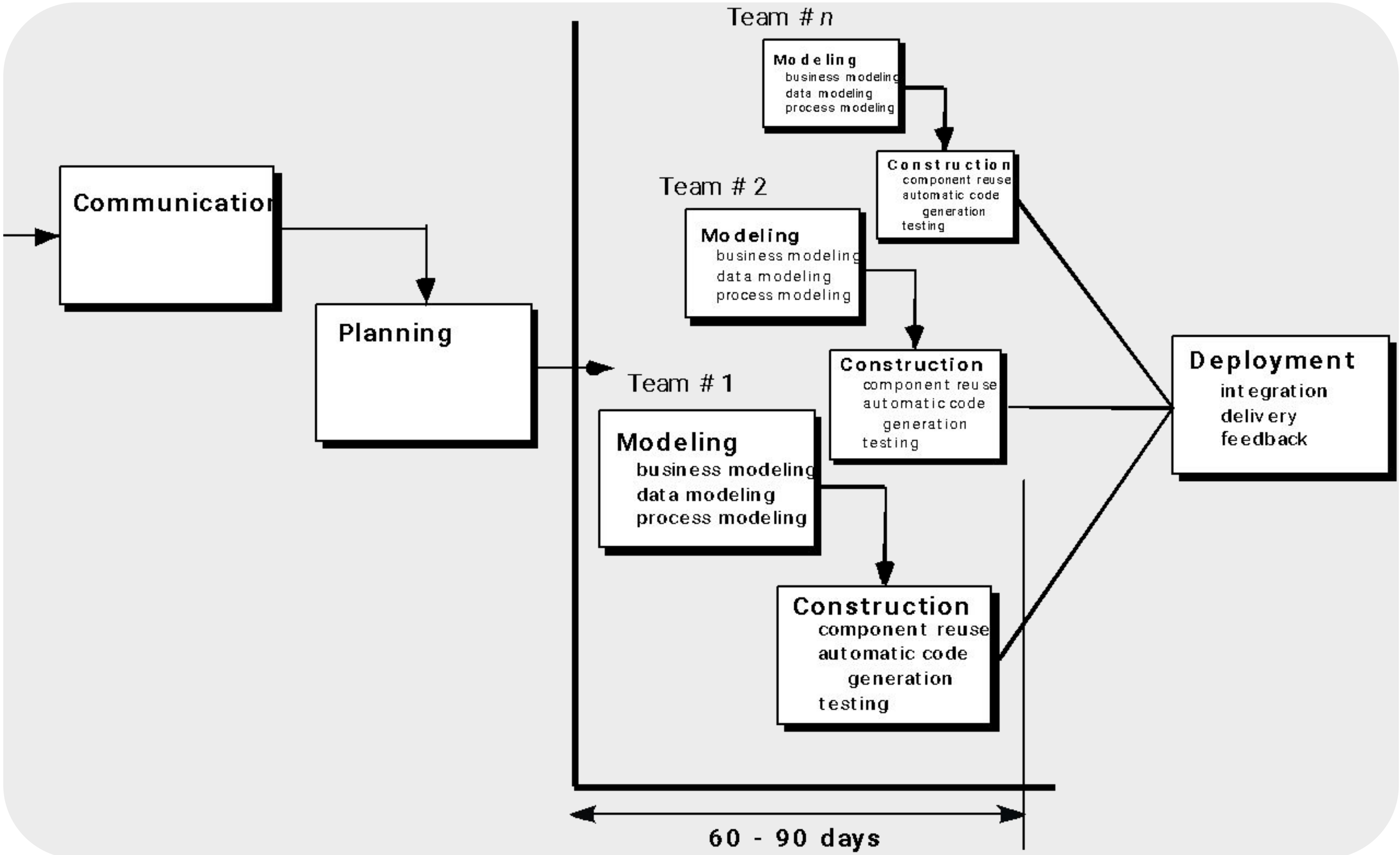


Figure : Flowchart of Incremental Model

The Rapid Application Development (RAD) Model



RAD Model

- It is type of incremental process model in which there is **extremely short development cycle.**
- When the requirements are fully understood and the **component based construction** approach is adopted then the RAD model is used.
- Using RAD model fully functional system can be developed within **60 to 90 days**.
- Multiple teams work on developing the s/w system using RAD model parallely .
- During design phase various models are created. Those models are Business model ,data model and process model.

Drawbacks of RAD model:

- It requires multiple teams or large number of people to work on the project.
- This model requires heavily committed developer and customers. If commitment is lacking then RAD projects will fail.
- Project using RAD models requires heavy resources.

When to choose incremental model?

- When requirements are reasonably well_defined.
- When overall scope of development effort suggest a purely linear effort.
- When limited set of s/w functionality needed.

Merits of Incremental model:

- The incremental model can be adopted when there are less number of people involved in the project.
- Technical risk can be managed with each increment.
- For a very small time span at least core product Can be delivered to client.

Questions

- What are the elements of Waterfall model? State its merits and demerits.
- Explain with neat diagram incremental model and state its advantages and disadvantages.
- Explain RAD model with the help of diagram.

MCQ

1. Maintenance is the final phase in waterfall model

- A. True
- B. False

2. A stage in which individual components are integrated and ensured that they are error free to meet customer requirement.

a. Coding
b. Testing
c. Design
d. implementation

MCQ

- 3. The customer requirements are broken down into logical modules for ease of _____.
- A. inheritance
- B. design
- C. editing
- D. implementation

4. A step in waterfall model that involves a meeting with the customer to understand the requirement.

a. Requirement gathering
b. SRS
c. Implementation
d. Customer review

Evolutionary process model

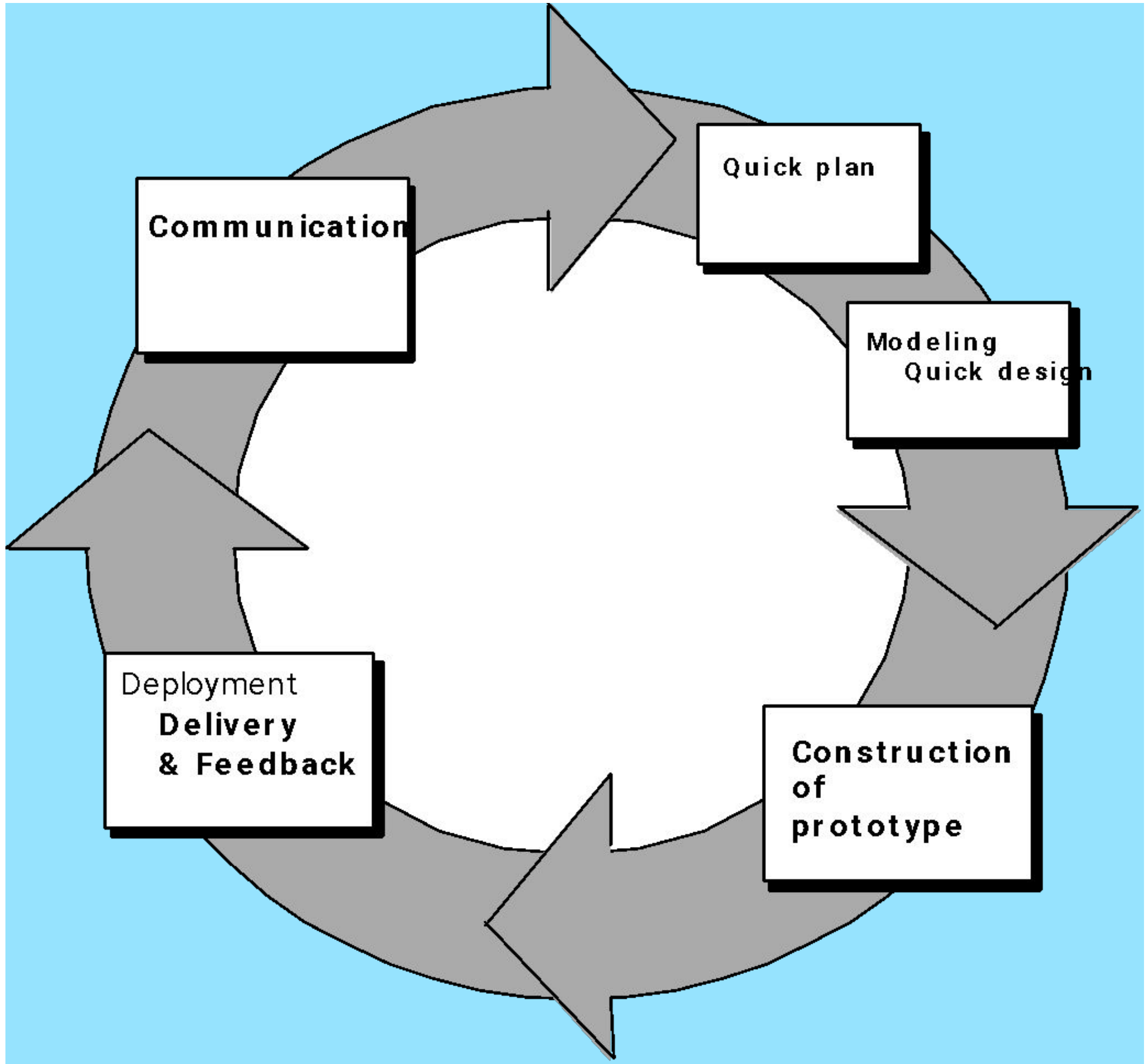
[pg no 42-46 PR]

- While developing the s/w system it is often needed to make modification in earlier development phases or task sets.
- IF development process is linear or in a straight line then the end product will be unrealistic.
- In such cases the **iterative approach** needs to be adapted.
- Evolutionary model is **iterative model**.

Prototyping:

- Software prototyping refers to building s/w application prototypes which displays the functionality of the product under development, but may not actually hold the exact logic of the original s/w.
- From a quick design a prototype is prepared.
- Customer or user evaluates the prototype in order to refine requirement .Iteratively prototype is tuned for satisfying customer requirements.

Evolutionary Models: Prototyping



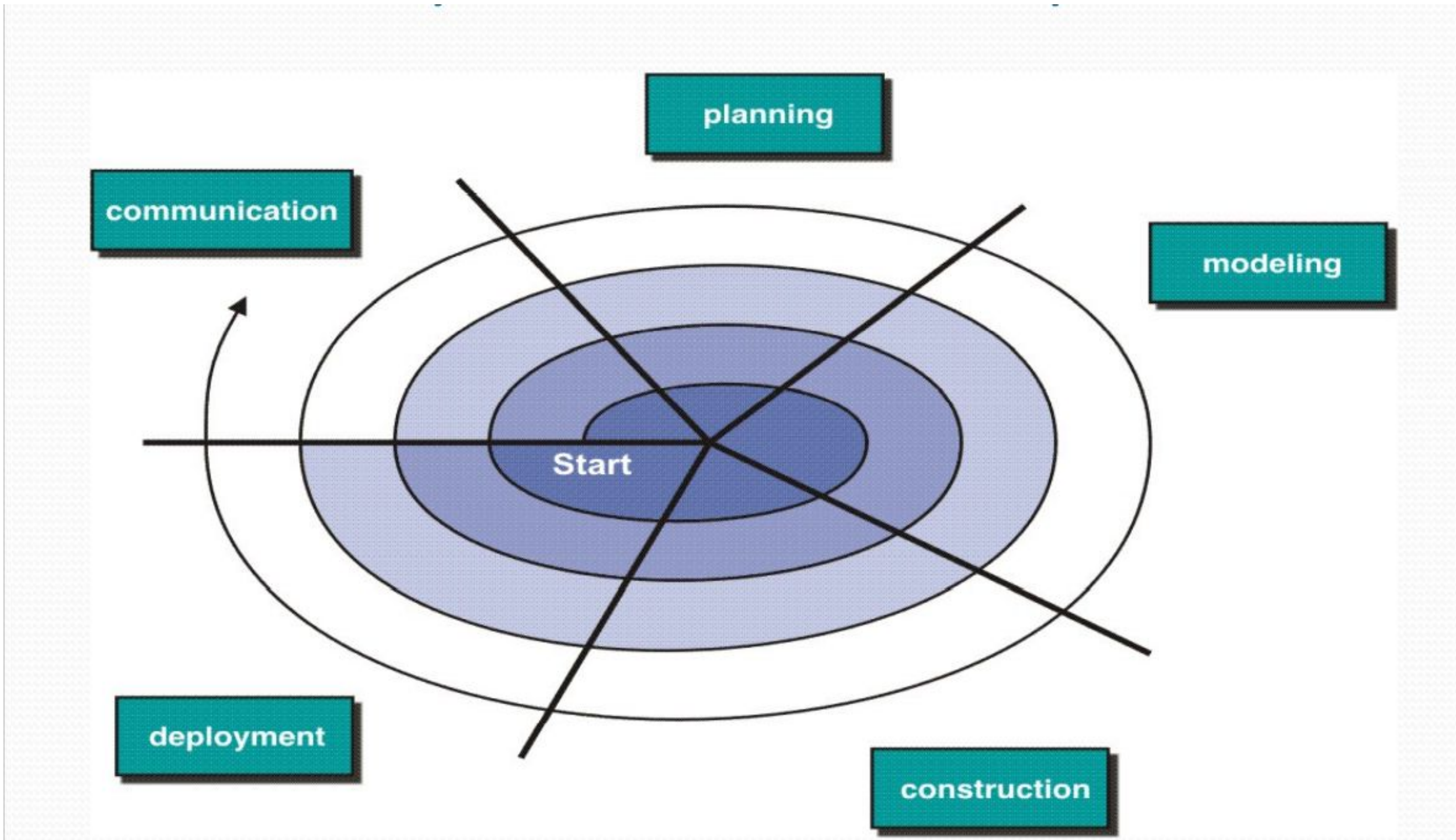
When to choose it?

- General objective of s/w is defined but not detailed input, processing or o/p requirement.
- When the developer is unsure about the efficiency of an algorithm then prototype serves as a better choice.

Drawback of prototyping:

- In the first version itself customer often want “few fixes” rather than rebuilding of the system.
- The first version may have some compromises.
- Sometimes developer may make implementation compromises to get prototype working quickly.

Evolutionary process model
(Spiral Model) [pg no 45-47]



- This model gives efficient development of incremental versions of s/w.
- In this model the s/w is developed in series of increments.
- Spiral model is divided into number of framework activities.
- These framework activities are denoted by task regions.
- Usually there are 6 task regions.
- Spiral model is realistic approach for development of **large scale systems** and software.
- Customer and developer identify risk in each level.

Advantages:

- Requirement changes can be made at every stage.
- Risk can be identified and rectified before they get problematic.

Drawbacks:

- If communication is not proper then s/w development will not be upto the mark.
- IF risk assessment is not done properly then product will not be obtained successfully.

Concurrent Development model

[pg.no.48 PR. 7th edition]

- It is also called as **concurrent Engineering.**
- In this model framework activities or software development tasks are representation as **states**.
- The modeling or designing phase of s/w development can be one of the states like under development waiting for modification under revision or under review and so on
- All the s/w development activities exist concurrently in this model but framework activities can be in various states.
- These states make transitions.

- i.e. during modeling the transition from under development states to waiting for modification state occurs.
- This model basically defines the series of events due to transition from one state to another state occurs

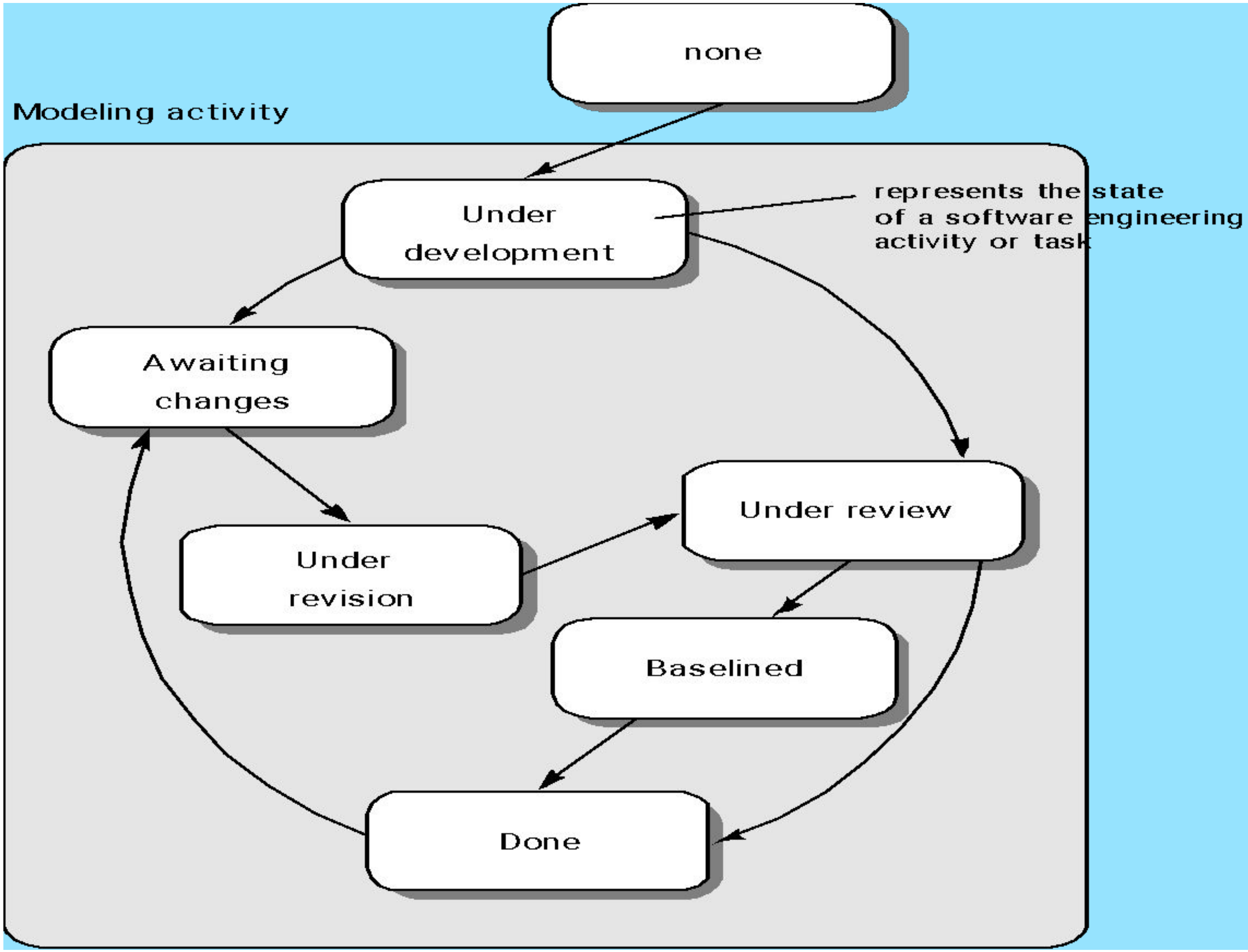


Figure. Concurrent development model

Advantages:

- All types of s/w development can be done using concurrent development model.
- This model provide accurate picture of current state of project.
- Each activity or tasks can be carried out concurrently.
- Hence this model is an efficient process model.

Specialized Process Models

- Specialized process models are used when only collection of specialized technique or methods are expected for developing specific s/w.
- Various types of specialized models are:

1. Component Based Development
2. Formal Methods Model
3. Aspect-Oriented Software Development

1.Component Based Development:

- These components have specialized targeted functionalities and well defined interface.
- It uses various characteristics of spiral models.
- This model is evolutionary in nature.
- The component must be searched and analyzed before beginning of modeling and construction activity of s/w development.
- Components can be simple functions or object oriented methods or classes.

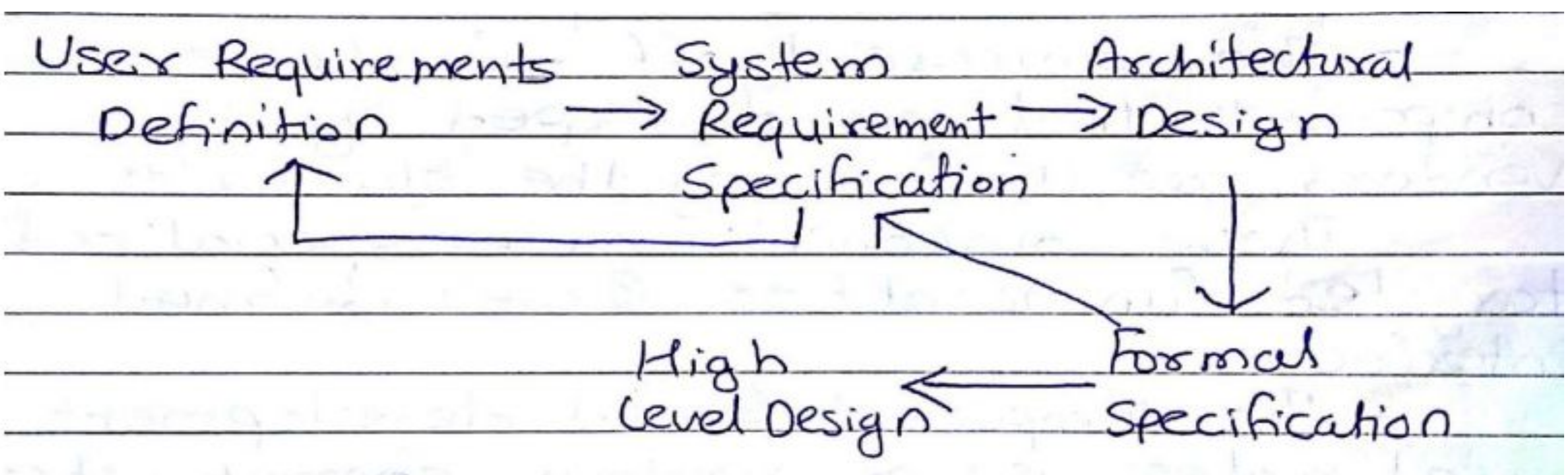
- **Steps:**

1. Identify component based product.
2. Analyse component integration issues.
3. Design s/w architecture(SA) to accommodate the component.
4. Integrate components into the SA.
5. Conduct comprehensive testing for the s/w.

- s/w reusability is the major advantage .
- Reusability reduces the development cycle time and overall cost.

2.Formal method model:[pg.no 51-53]

- This model consist of set of activities in which the formal mathematical specification is used.
- The s/w engg. Specify, develop and test the computer based systems using mathematical notation.
- The notations are specified within the formal method.
- Clean room s/w engineering makes use of the formal method approach.



Advantages:

- Ambiguity
- Incompleteness
- Inconsistency
- Offers defect free s/w

Drawbacks:

- It is time consuming and expensive.
- For using this model developer must have strong mathematical background or some extensive training.
- If this model is chosen for development then communication with customer become very difficult.

3.Aspect-Oriented Software Development:

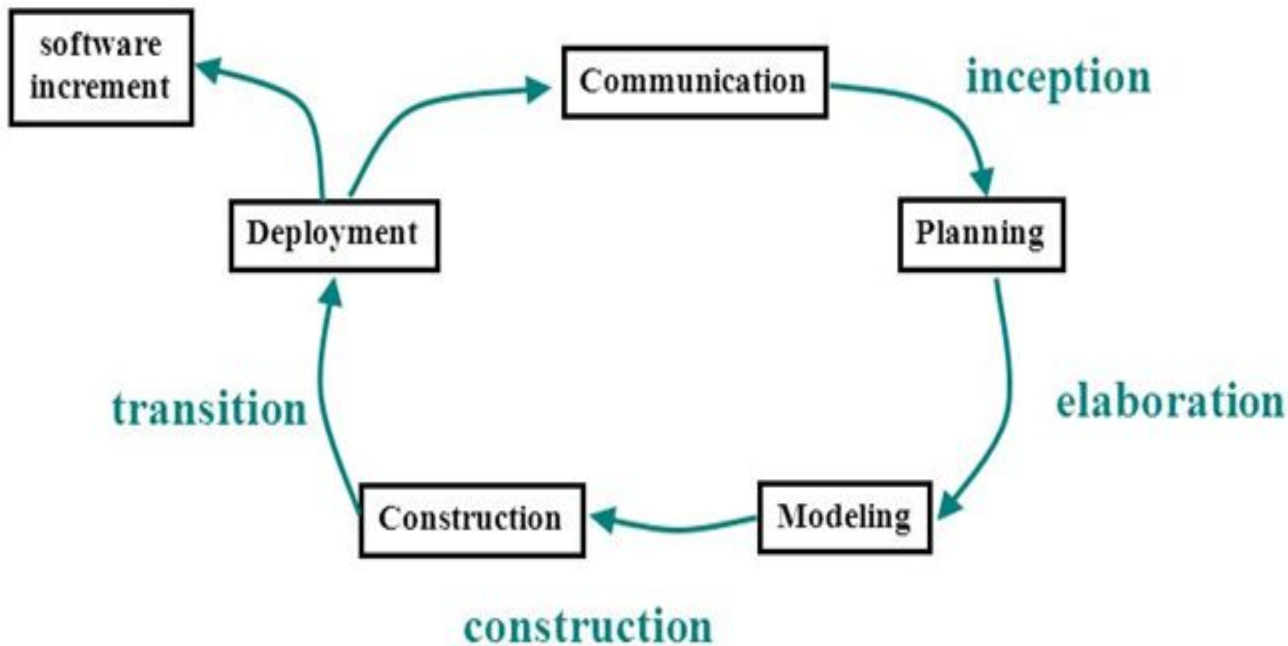
- This focuses on identification,specification and representation of cross cutting concern and their modularization into separate functional unit.
- Aspectual requirements defines these cross cutting concerns that have impact on s/w architecture.
- It is relatively new s/w engg. Paradigm and is not matured enough.

The Unified process

- IT is a framework for object oriented model.
- This model is also called as rational unified process model(RUP).
- This model is **iterative** and **incremental** by nature.
- There are phases:

1. Inception
2. Elaboration
3. Construction
4. Transition
5. Production

Unified process



1.Inception: Two major activities communication and planning

- Initial use cases are created.
- Initial risk assessment
- Project planning

2.Elaboration:

- Planning and modeling
- Use cases are redefined
- Architectural representation is created using five models such as-

i)use case model

ii)Analysis model

iii) Design model

iv)Implementation model

v) Deployment model

3.Construction :

- Main activity is to make use cases operational.
- Analysis and design activities of previous phases are completed and source code is developed.
- Then testing and user and installation manual is created.

4.Transition:

- Activities required for deployment of s/w product are carried out.
- Beta testing conducted.
- User feedback report is used to remove defects from created system.

5.Production:

- Mainly maintenance activities are conducted in order to support the user in user friendly environment.

Unit :1

Software Engineering Fundamentals

-Software Myths

-A Process Framework

-Process Models
-Waterfall
-Incremental
-RAD
-Spiral
-Prototype

Advanced Process Models & Tools

-Agile Modeling
-XP
-Scrum

-Tools
-JIRA
-Kanban


Agile

What Is Agile

Meaning: able to move quickly and easily.

- Readiness for motion of activity.
- Effective response to change.
- Effective communication among all stakeholder.
- Organizing a team so that it is in control of the work performed.
- Rapid, Incremental delivery of s/w.

Agility

- The ability to both create and respond to change in order to profit in a business environment.
- Agile engineers believe:

✔ Current s/w process are heavyweight.

✔ Too many things are done that are not directly related to s/w product being produced.

✔ Current s/w development is too rigid.

✔ Difficulty with incomplete or changing requirements.

✔ Short development cycles.(internet applications)

✔ More active costumer involvement needed.

✔ CMM(capability maturity model) focuses on process.

- **Agile methods are considered**:

 Lightweight

 People based rather than plan based.

 Set of principles

 Developed by agile alliance -

- A non profit organization promotes agile development.
- agile is dynamic, content specific, aggressively, change embracing and growth oriented.

- **An Agile process –**
- **Characteristics**

1. **Adaptable to technical & environmental changes**
2. **Development is incremental**
3. **Customer feedback is must**
4. **Delivered in short span of time**
5. **Iterative in nature, so evaluated regularly**

Merits

1. **Customer satisfaction**
2. **Easy interaction**
3. **Continuous attention**
4. **Lates changes easily accommodated**

Agile process

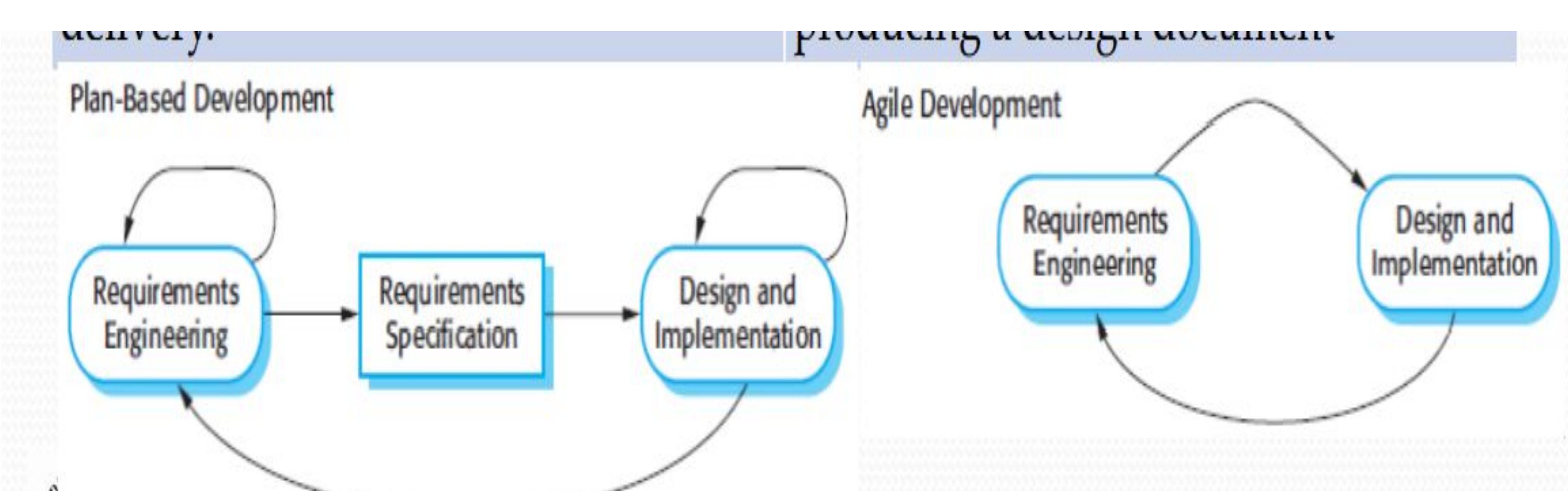
- Demerits
- 1. lack of designing & documentation
- 2. easily get off track

Agile principles

1. Highest priority is to satisfy the customer through early and continuous delivery of valuable s/w.
2. Welcome changes from customer at anything in the process.
3. Deliver working s/w frequently.
4. Business people and developer must work together daily throughout the project.
5. Build project around motivated individual.
6. Have face to face conversation.
7. Working s/w is primary measure of progress.
8. The sponsors, developers and user should be able to maintain a constant pace indefinitely.
9. Continuous attention to technical excellence and good designs enhances agility.
10. simplicity.

Plan driven and agile development

Plan Driven Approach	Agile Approach
Plan Driven Approach identifies separate stages in the software process with outputs associated with each stage	Agile approach consider design and implementation to be the central activities in the software process
Iteration occurs within activities.	Iteration occurs across activities.
Supports incremental development and delivery.	Not code-focused and end up producing a design document



Requirements change request

Success in industry

The success of project, which follow extreme programming practices, is due to -

- Rapid development
- Immediate responsiveness to the customer’s changing requirements.
- Focus on low defect rates
- System returning constant and consistent value to the customer.
- High customer satisfaction.
- Reduced cost.
- Team cohesion and employee satisfaction.

Agile methods

- Scrum
- Extreme Programming
- Adaptive s/w development (ASD)
- Dynamic system development method(DSDM)etc.

Agile methodologies

- **Kanban** - Kanban is a simple, visual means of managing projects that emphasizes visibility.
- **Scrum** - Scrum focuses on breaking a project down into sprints, and only planning and managing one sprint at a time. Scrum also has unique project roles, including a scrum master and Product Owner.
- **Extreme Programming(XP)** - It focuses on continuous development and customer delivery and uses intervals or sprints similar to a scrum methodology

Feature-Driven Development (FDD) - This methodology involved creating software models every two weeks and requires a development and design plan for every software model feature. Therefore, it has more rigorous documentation requirements than XP.

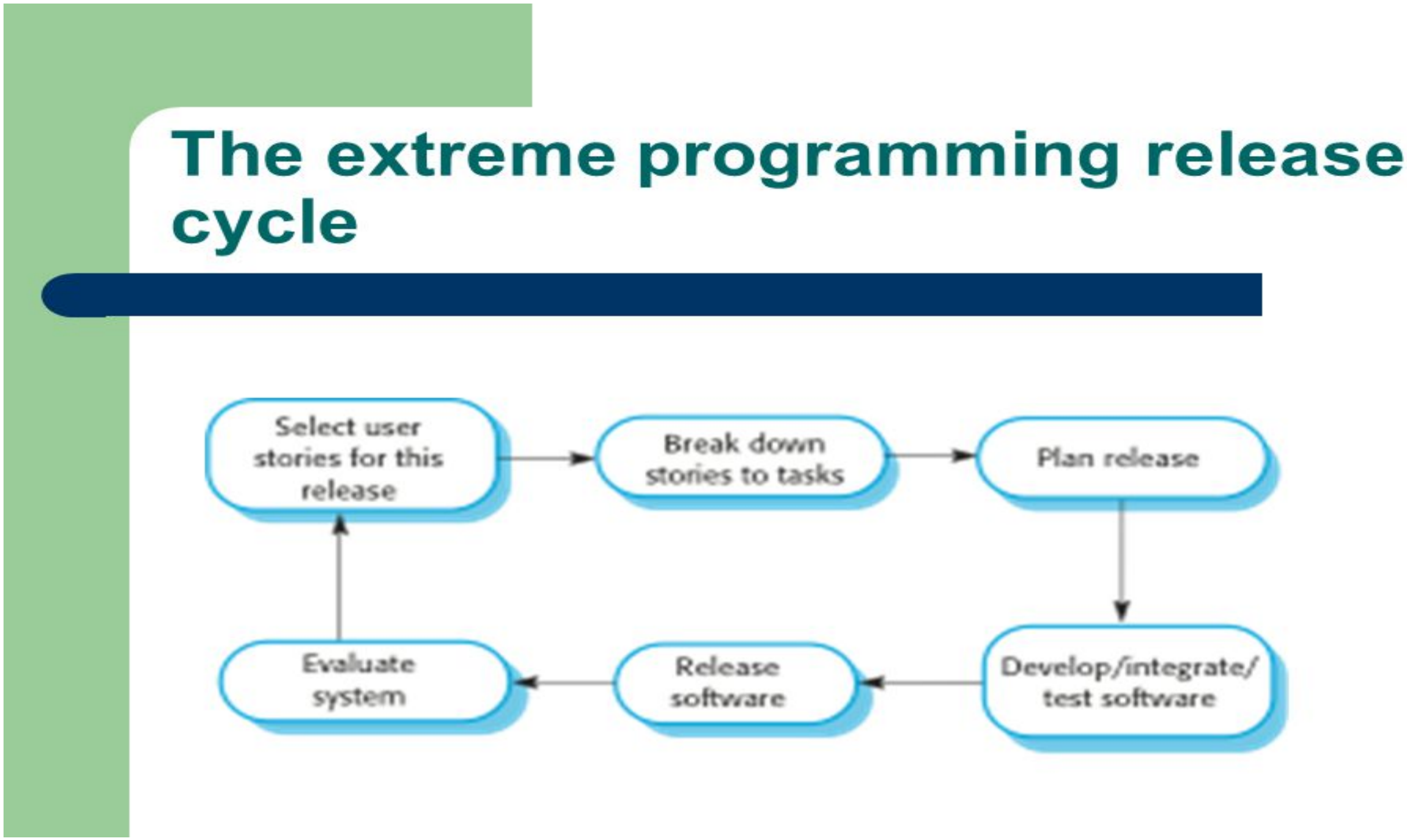
FDD breaks projects down into five basic activities:

- Develop an overall model
- Build a feature list
- Plan by feature
- Design by feature
- Build by feature

Dynamic Systems Development Method (DSDM) - The Dynamic Systems Development Method (DSDM) cropped up out of the need to provide a common industry framework for rapid software delivery. Part of DSDM is the mandate that rework is to be expected, and any development changes that occur must be reversible.This framework is based on eight key principles:

- Focus on the business need
- Deliver on time
- Collaborate
- Never compromise quality
- Build incrementally from firm foundations
- Develop iteratively
- Communicate continuously and clearly
- Demonstrate control

Extreme Programming



XP Practices

- Incremental Planning
- Small releases
- Simple design
- Test first development
- Refactoring
- Pair programming
- Collective Ownership
- Continuous Programming

Questions

- Describe agile in detail.
- What is agility? Explain about process model.
- What is XP methology for project management?

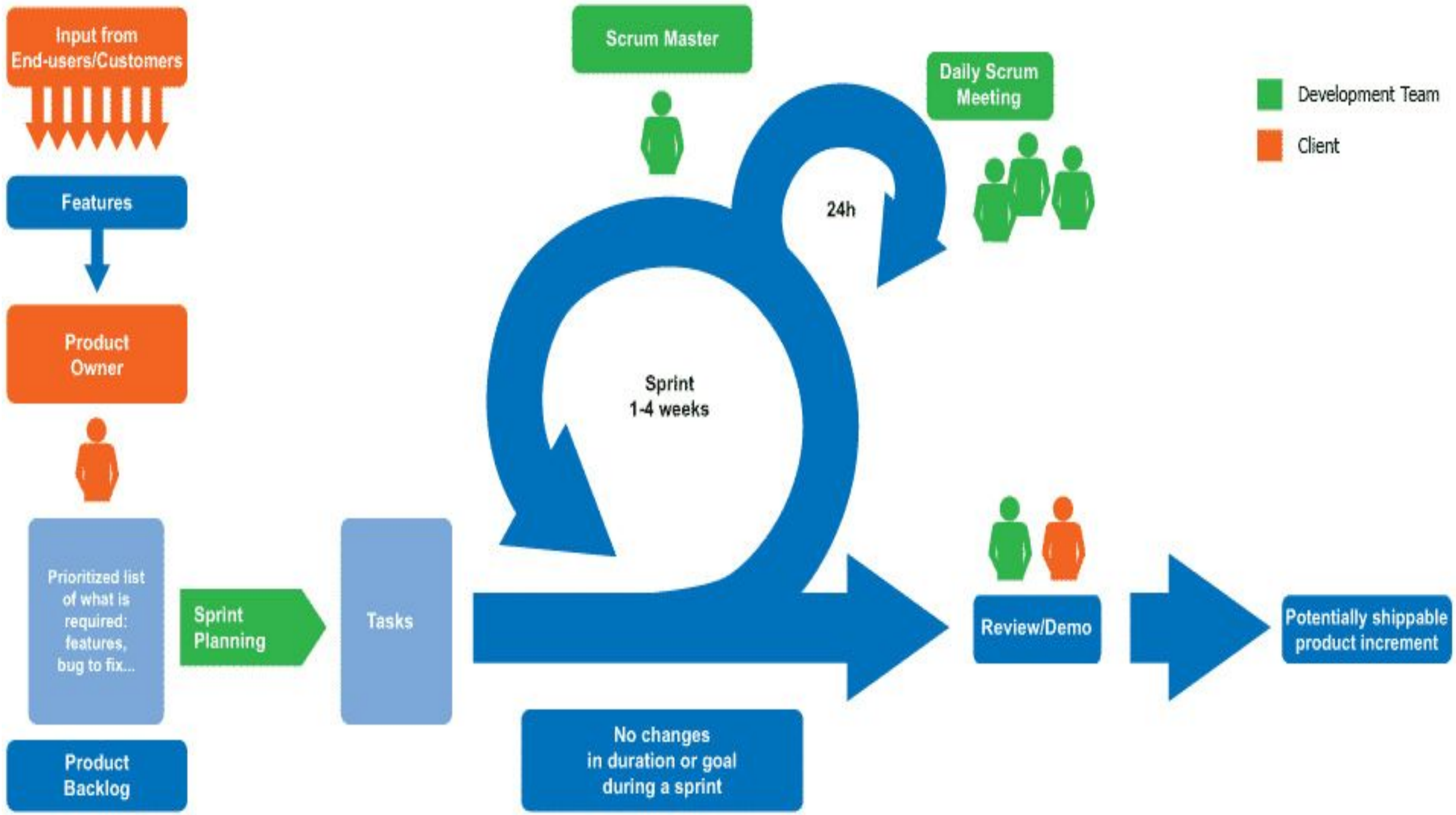
MCQ

- Agile Software Development is based on
  a) Incremental Development
  b) Iterative Development
  c) Linear Development
  d) Both Incremental and Iterative Development

- Incremental development in Extreme Programming (XP) is supported through a system release once every month.
  a) True
  b) False

- User requirements are expressed as __________ in Extreme Programming.
  a) implementation tasks
  b) functionalities
  c) scenarios
  d) none of the mentioned

Scrum


Input from
End-users/Customers
Features
Product
Owner
Prioritized list
of what is
required:
features,
bug to fix...
Product
Backlog
Sprint
Planning
Tasks
Scrum Master
Daily Scrum
Meeting
24h
Sprint
1-4 weeks
No changes
in duration or goal
during a sprint
Review/Demo
Potentially shippable
product increment
Development Team
Client

Scrum

- Used for developing the complex software
- Lightweight process framework
- Iterative and incremental

Principles

- Small working teams
- Packets or partitions
- Accommodate changes
- Software increments
- Testing & documentation
- Produce working model

Activities

- Backlog: list of project requirements
- Sprint: work unit
- Meetings: daily meetings
- Demo: increment delivery

Roles

- Scrum Master
- Team Member

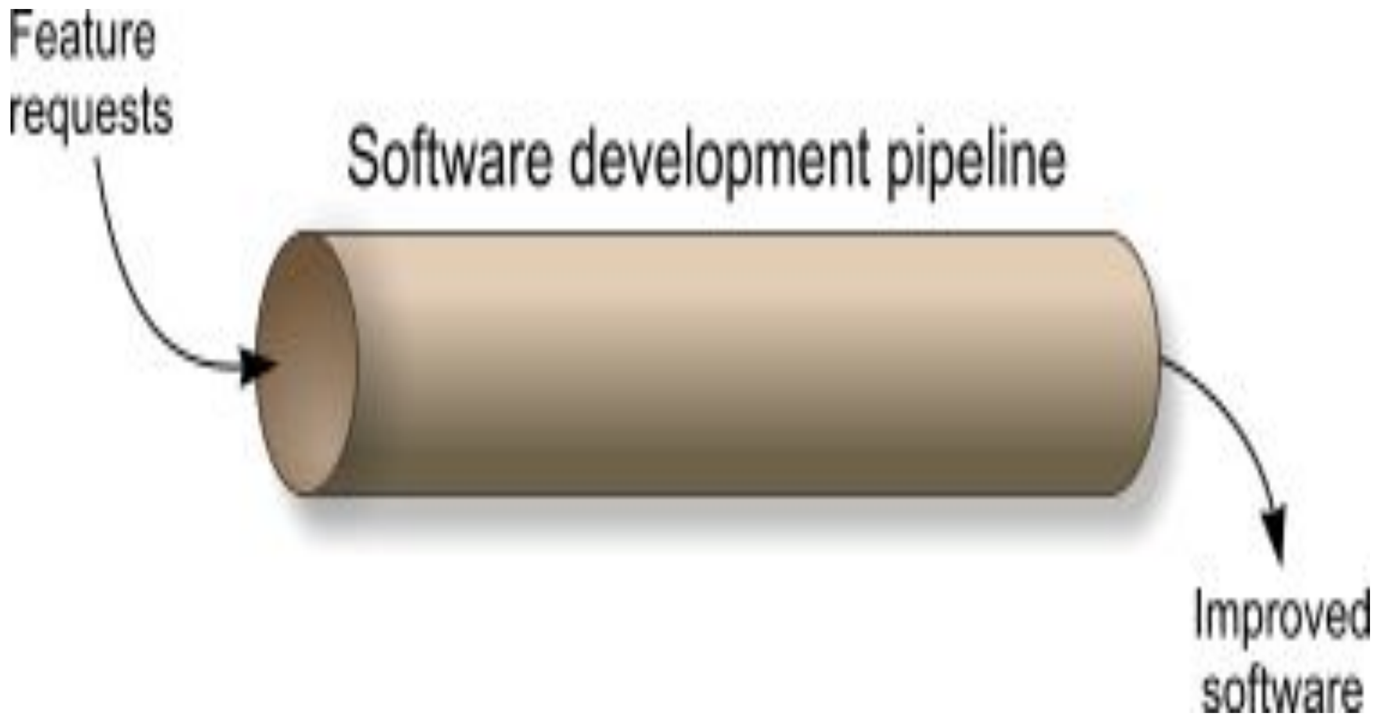
Advantages

- Transparency
- Flexibility
- Minimum overhead
- Productivity
- Disadvantages
- Hard to track decisions
- Non functional requirements

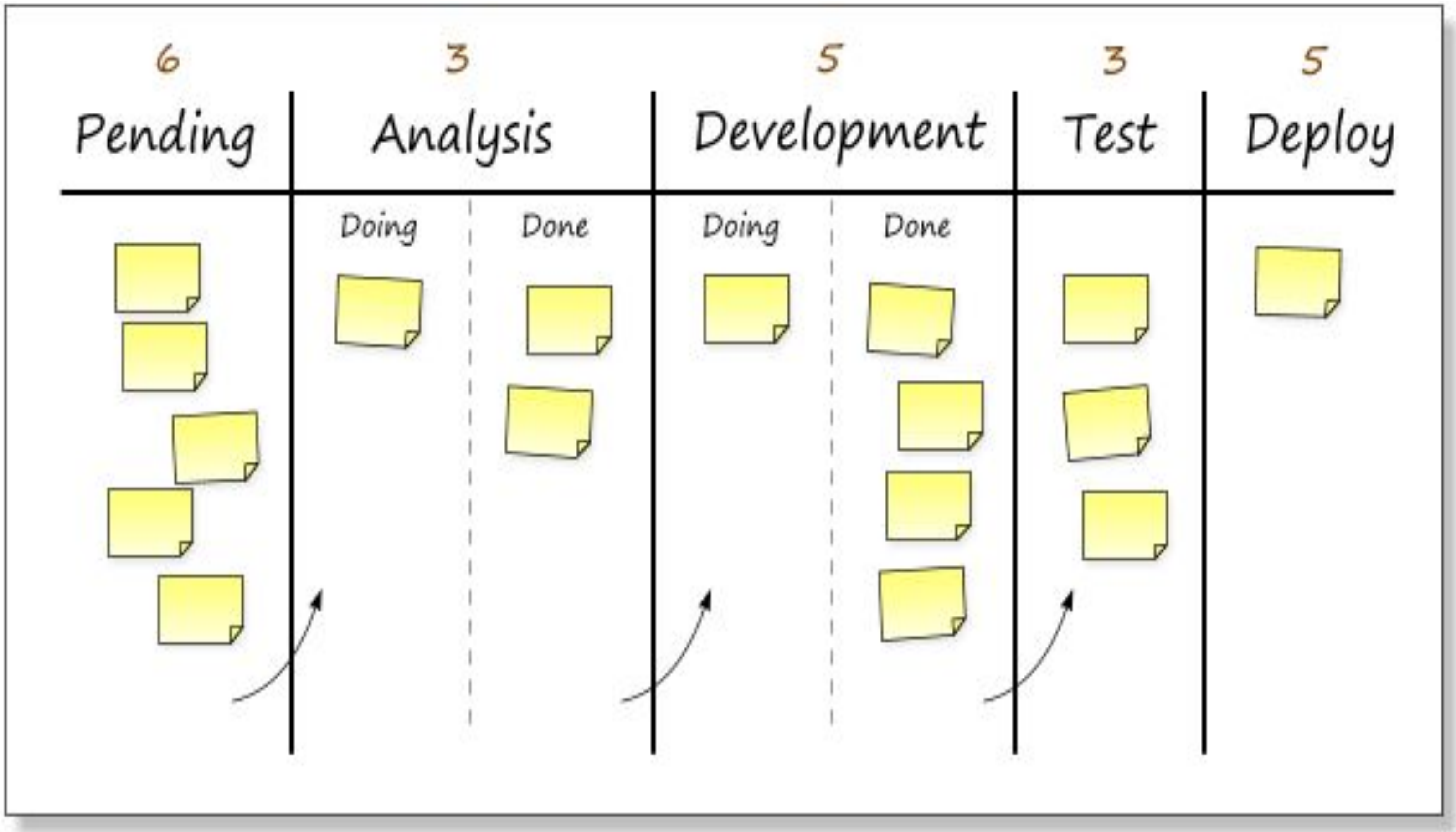
Introduction to agile tool

- The Objective is to help in one or more activities in agile development.
- Automated support for project planning, use case development, requirement gathering, code generation and testing.
- Project management tools focus on preparing —Earned Value Charts instead of preparing the Gantt Charts.
- Purpose is to enhance the environment

Kanban


Feature requests
Software development pipeline
Improved software

Kanban



6
Pending
3
Analysis
Doing
Done
5
Development
Doing
Done
3
Test
5
Deploy

JIRA

- Developed by Atlassian in 2002.
- Used for bug tracking, issue tracking and project management.
- Inherited from Japanese word —Gojira meaning Godzilla
- JIRA is a competitor for Bugzilla.
- Written in **Java** and used for **Project Management activities.**

Features:

- Plan development iterations
- Iteration reports
- Bug tracking

JIRA Issues:

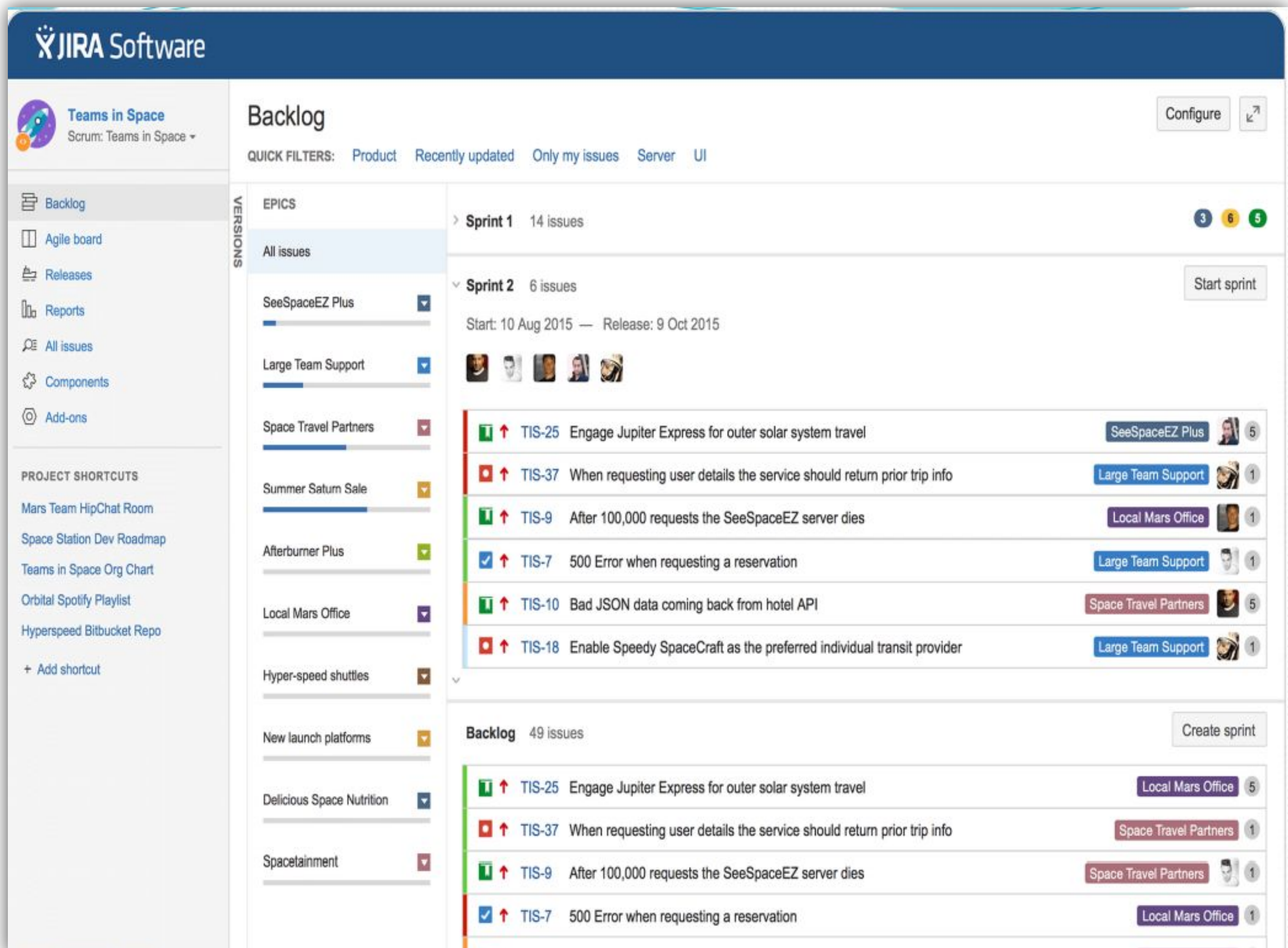
- It is a part of JIRA tool which is used to track bugs in the project.

JIRA Components :

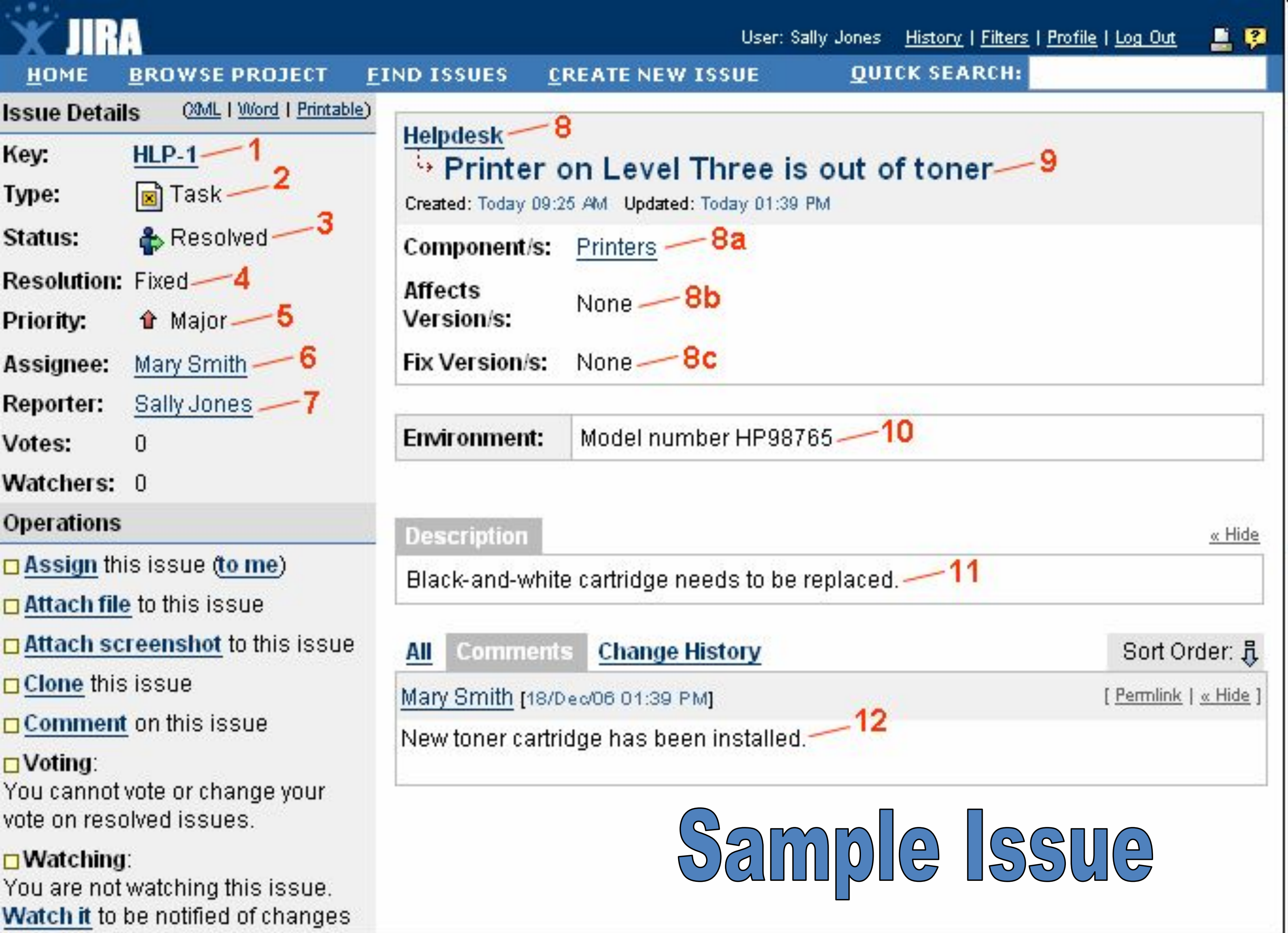
- Sub-sections of project
- Used to group issues into smaller parts.
- Projects can be broken down into features, teams, modules, sub-projects, etc.
- We can generate reports, collect statistics and display it on dashboards.

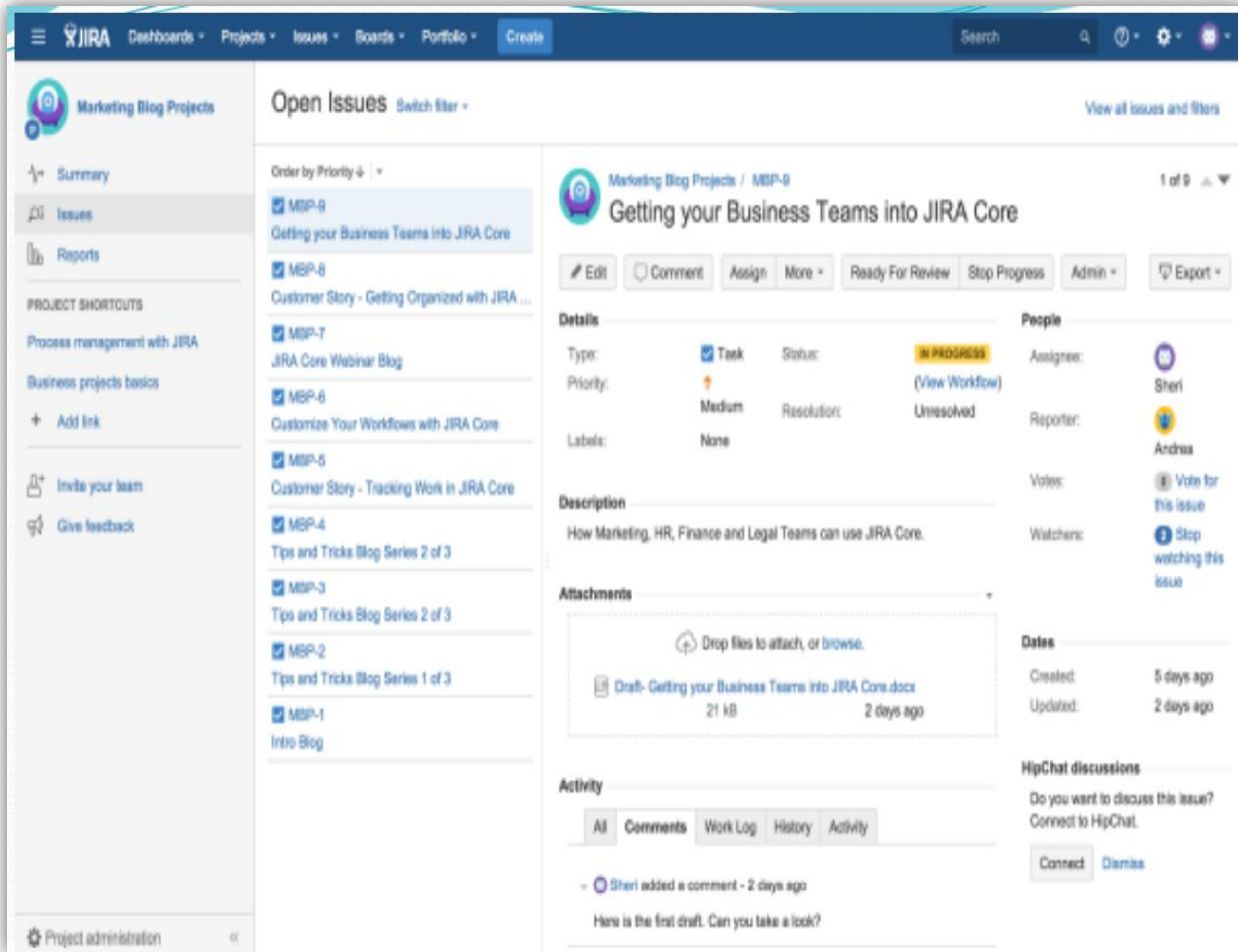
JIRA

- Reports in JIRA
- Progress of the project can be represented using BurndownChart.
- Different reports like Sprint Report, Version Report, Control Chart, Cumulative flow diagram, etc. are also available.


JIRA Software
Teams in Space
Scrum: Teams in Space
Backlog
Agile board
Releases
Reports
All issues
Components
Add-ons
PROJECT SHORTCUTS
Mars Team HipChat Room
Space Station Dev Roadmap
Teams in Space Org Chart
Orbital Spotify Playlist
Hyperspeed Bitbucket Repo
+ Add shortcut
Backlog
QUICK FILTERS: Product Recently updated Only my issues Server UI
Configure
VERSIONS
EPICS
All issues
SeeSpaceEZ Plus
Large Team Support
Space Travel Partners
Summer Saturn Sale
Afterburner Plus
Local Mars Office
Hyper-speed shuttles
New launch platforms
Delicious Space Nutrition
Spacetainment
Sprint 1 14 issues
3 6 5
Sprint 2 6 issues
Start sprint
Start: 10 Aug 2015 — Release: 9 Oct 2015
TIS-25 Engage Jupiter Express for outer solar system travel SeeSpaceEZ Plus 5
TIS-37 When requesting user details the service should return prior trip info Large Team Support 1
TIS-9 After 100,000 requests the SeeSpaceEZ server dies Local Mars Office 1
TIS-7 500 Error when requesting a reservation Large Team Support 1
TIS-10 Bad JSON data coming back from hotel API Space Travel Partners 5
TIS-18 Enable Speedy SpaceCraft as the preferred individual transit provider Large Team Support 1
Backlog 49 issues
Create sprint
TIS-25 Engage Jupiter Express for outer solar system travel Local Mars Office 5
TIS-37 When requesting user details the service should return prior trip info Space Travel Partners 1
TIS-9 After 100,000 requests the SeeSpaceEZ server dies Space Travel Partners 1
TIS-7 500 Error when requesting a reservation Local Mars Office 1

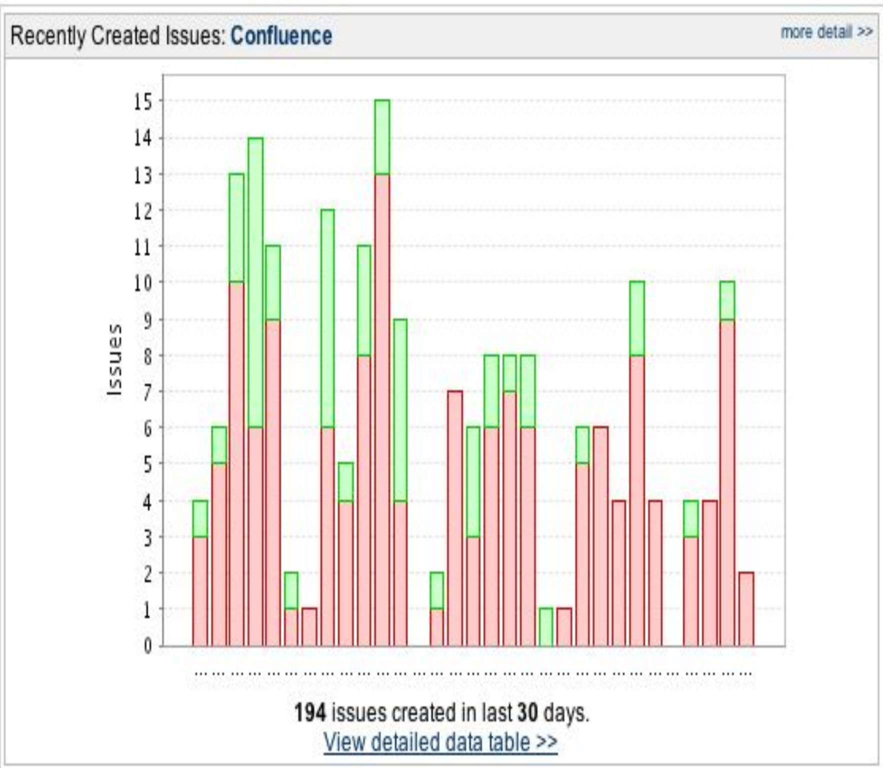
Sample Issue





JIRA
Dashboards
Projects
Issues
Boards
Portfolio
Create
Search
Marketing Blog Projects
Summary
Issues
Reports
PROJECT SHORTCUTS
Process management with JIRA
Business projects basics
Add link
Invite your team
Give feedback
Project administration
Open Issues
Switch filter
View all issues and filters
Order by Priority
MBP-9
Getting your Business Teams into JIRA Core
MBP-8
Customer Story - Getting Organized with JIRA ...
MBP-7
JIRA Core Webinar Blog
MBP-6
Customize Your Workflows with JIRA Core
MBP-5
Customer Story - Tracking Work in JIRA Core
MBP-4
Tips and Tricks Blog Series 2 of 3
MBP-3
Tips and Tricks Blog Series 2 of 3
MBP-2
Tips and Tricks Blog Series 1 of 3
MBP-1
Intro Blog
Marketing Blog Projects / MBP-9
1 of 9
Getting your Business Teams into JIRA Core
Edit
Comment
Assign
More
Ready For Review
Stop Progress
Admin
Export
Details
Type:
Task
Status:
IN PROGRESS
Priority:
(View Workflow)
Medium
Resolution:
Unresolved
Labels:
None
People
Assignee:
Sheri
Reporter:
Andrea
Votes:
Vote for this issue
Watchers:
Stop watching this issue
Description
How Marketing, HR, Finance and Legal Teams can use JIRA Core.
Attachments
Drop files to attach, or browse.
Draft- Getting your Business Teams into JIRA Core.docx
21 kB
2 days ago
Dates
Created:
5 days ago
Updated:
2 days ago
HipChat discussions
Do you want to discuss this issue?
Connect to HipChat.
Connect
Dismiss
Activity
All
Comments
Work Log
History
Activity
Sheri added a comment - 2 days ago
Here is the first draft. Can you take a look?

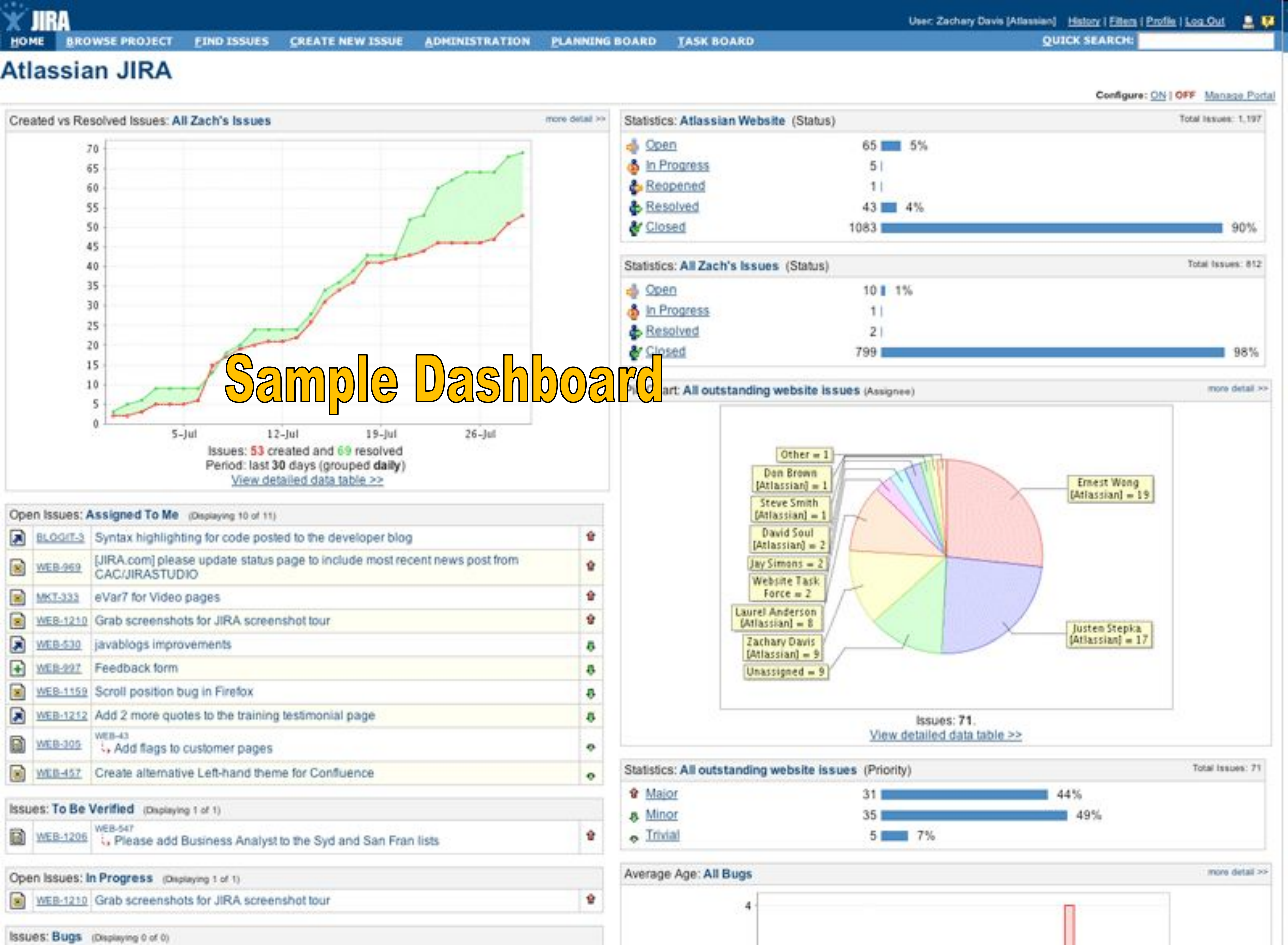
Statistics: **JIRA** (Issue Type)		
Bug	4539	45%
Improvement	2711	27%
New Feature	1351	13%
Sub-task	183	2%
Support Request	766	8%
Task	485	5%
Third-party issue	31	



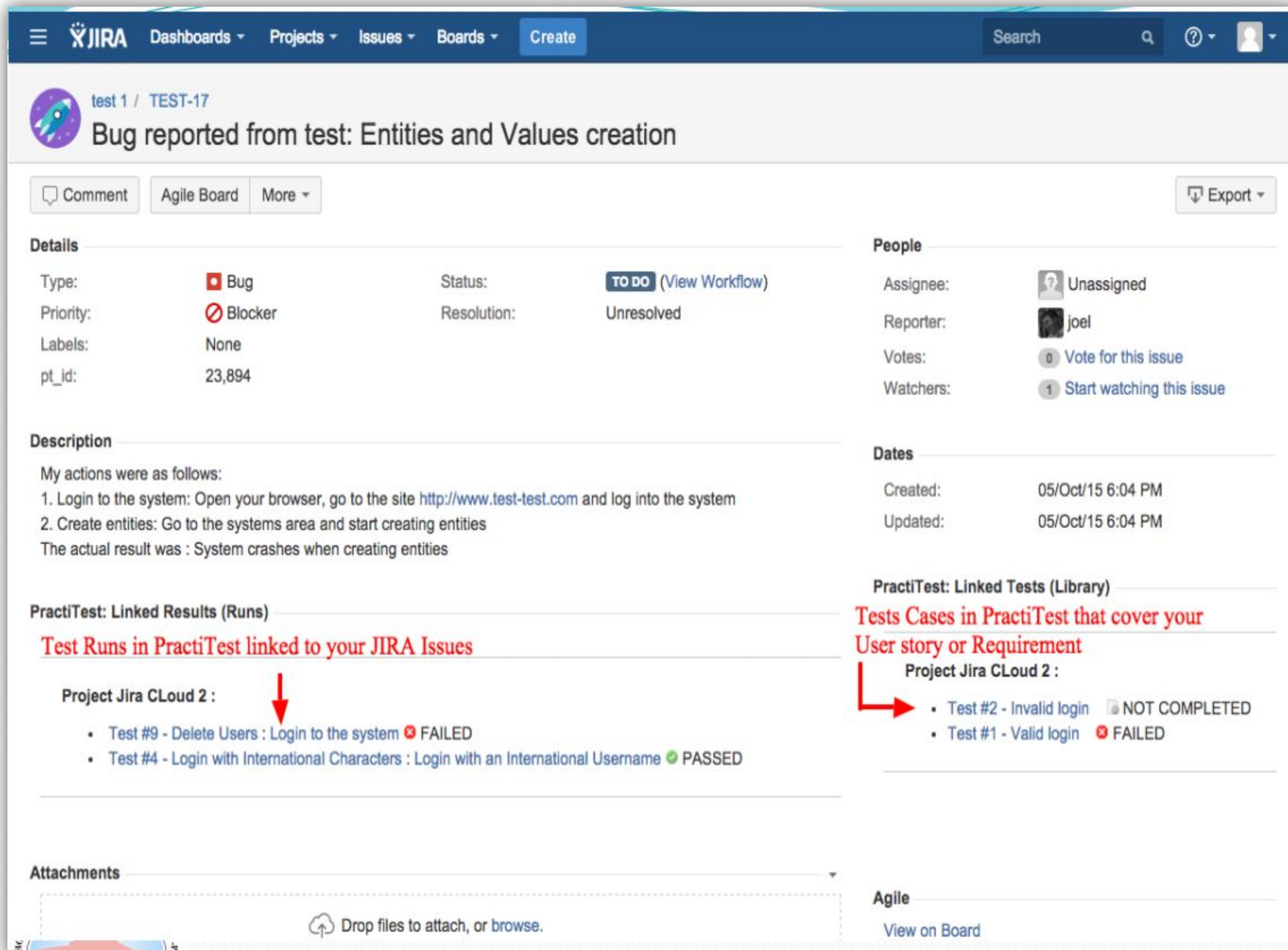
Statistics Table: JIRA's Last 30 Days Resolved Issues

Assignee / Priority	Critical	Major	Minor	Trivial	T:
Anton Mazkovoi	0	2	0	0	2
Chris Mountford	0	4	1	0	5
Daniel Hurst	0	8	8	2	18
Dushan Hanuska	0	3	0	0	3
Dylan Etkin	2	8	0	0	10
Jed Wesley-Smith	0	2	0	0	2
Jeff Turner	0	1	0	0	1
Justin Koke	2	5	0	2	9
Keith Brophy	1	0	0	0	1
Mark Chaimungkalanont	1	0	0	0	1
Total Unique Issues:	**15**	**92**	**28**	**8**	**143**

Sample Reports - II



JIRA
User: Zachary Davis [Atlassian] History | Filters | Profile | Log Out
HOME BROWSE PROJECT FIND ISSUES CREATE NEW ISSUE ADMINISTRATION PLANNING BOARD TASK BOARD QUICK SEARCH:
Atlassian JIRA
Configure: ON | OFF Manage Portal
Created vs Resolved Issues: All Zach's Issues
more detail >>
Issues: 53 created and 69 resolved
Period: last 30 days (grouped daily)
View detailed data table >>
Statistics: Atlassian Website (Status)
Total Issues: 1,197
Open 65 5%
In Progress 5
Reopened 1
Resolved 43 4%
Closed 1083 90%
Statistics: All Zach's Issues (Status)
Total Issues: 812
Open 10 1%
In Progress 1
Resolved 2
Closed 799 98%
Sample Dashboard
All outstanding website issues (Assignee)
Other = 1
Don Brown [Atlassian] = 1
Steve Smith [Atlassian] = 1
David Soul [Atlassian] = 2
Jay Simons = 2
Website Task Force = 2
Laurel Anderson [Atlassian] = 8
Zachary Davis [Atlassian] = 9
Unassigned = 9
Ernest Wong [Atlassian] = 19
Justen Stepka [Atlassian] = 17
Issues: 71.
View detailed data table >>
Open Issues: Assigned To Me (Displaying 10 of 11)
BLOGIT-3 Syntax highlighting for code posted to the developer blog
WEB-969 [JIRA.com] please update status page to include most recent news post from CAC/JIRASTUDIO
MKT-333 eVar7 for Video pages
WEB-1210 Grab screenshots for JIRA screenshot tour
WEB-530 javablogs improvements
WEB-997 Feedback form
WEB-1159 Scroll position bug in Firefox
WEB-1212 Add 2 more quotes to the training testimonial page
WEB-305 WEB-43 Add flags to customer pages
WEB-457 Create alternative Left-hand theme for Confluence
Issues: To Be Verified (Displaying 1 of 1)
WEB-1206 WEB-547 Please add Business Analyst to the Syd and San Fran lists
Open Issues: In Progress (Displaying 1 of 1)
WEB-1210 Grab screenshots for JIRA screenshot tour
Issues: Bugs (Displaying 0 of 0)
Statistics: All outstanding website issues (Priority)
Total Issues: 71
Major 31 44%
Minor 35 49%
Trivial 5 7%
Average Age: All Bugs
more detail >>


JIRA
Dashboards
Projects
Issues
Boards
Create
Search
test 1 / TEST-17
Bug reported from test: Entities and Values creation
Comment
Agile Board
More
Export
Details
Type: Bug
Priority: Blocker
Labels: None
pt_id: 23,894
Status: TO DO (View Workflow)
Resolution: Unresolved
People
Assignee: Unassigned
Reporter: joel
Votes: 0 Vote for this issue
Watchers: 1 Start watching this issue
Description
My actions were as follows:
1. Login to the system: Open your browser, go to the site http://www.test-test.com and log into the system
2. Create entities: Go to the systems area and start creating entities
The actual result was : System crashes when creating entities
Dates
Created: 05/Oct/15 6:04 PM
Updated: 05/Oct/15 6:04 PM
PractiTest: Linked Results (Runs)
Test Runs in PractiTest linked to your JIRA Issues
Project Jira CLoud 2 :
Test #9 - Delete Users : Login to the system FAILED
Test #4 - Login with International Characters : Login with an International Username PASSED
PractiTest: Linked Tests (Library)
Tests Cases in PractiTest that cover your User story or Requirement
Project Jira CLoud 2 :
Test #2 - Invalid login NOT COMPLETED
Test #1 - Valid login FAILED
Attachments
Drop files to attach, or browse.
Agile
View on Board

Questions

- Write a note on Kanban.
- Explain Scrum in detail.

Principles behind the Agile Manifesto

- Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
- Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
- Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
- Business people and developers must work together daily throughout the project.

- Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
- The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
- Working software is the primary measure of progress.
- Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

- Continuous attention to technical excellence and good design enhances agility.
- Simplicity--the art of maximizing the amount of work not done--is essential.
- The best architectures, requirements, and designs emerge from self-organizing teams.
- At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

The Agile Manifesto

- We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:
  - Individuals and interactions over processes and tools
  - Working software over comprehensive documentation
  - Customer collaboration over contract negotiation
  - Responding to change over following a plan

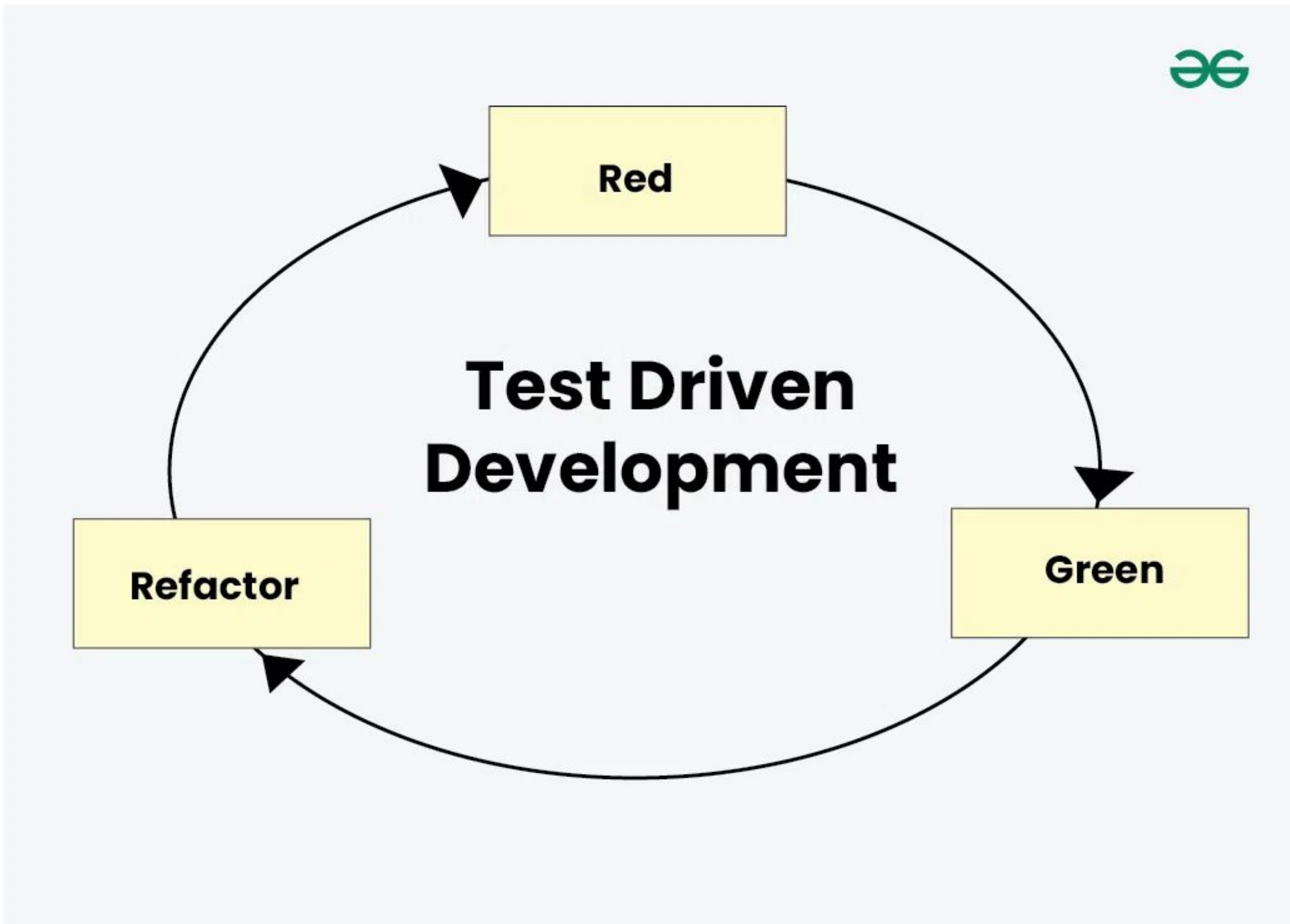
History: The Agile Manifesto

- On February 11-13, 2001, at The Lodge at Snowbird ski resort in the Wasatch mountains of Utah, seventeen people met to talk, ski, relax, and try to find common ground—and of course, to eat.
- What emerged was the Agile 'Software Development' Manifesto.
- Representatives from Extreme Programming, SCRUM, DSDM, Adaptive Software Development, Crystal, Feature-Driven Development, Pragmatic Programming, and others sympathetic to the need for an alternative to documentation driven, heavyweight software development processes convened.
- Now, a bigger gathering of organizational anarchists would be hard to find, so what emerged from this meeting was symbolic—a Manifesto for Agile Software Development—signed by all participants.
- The only concern with the term agile came from Martin Fowler (a Brit for those who don't know him) who allowed that most Americans didn't know how to pronounce the word 'agile'.

Test Driven Development

- TDD is a software development methodology that emphasizes writing tests before writing the actual code.
- It ensures that code is always tested and functional, reducing bugs and improving code quality.
- In TDD, developers write small, focused tests that define the desired functionality, then write the minimum code necessary to pass these tests, and finally, refactor the code to improve structure and performance.
- This cyclic process helps in creating reliable, maintainable, and efficient software. By following TDD, teams can enhance code reliability, accelerate development cycles, and maintain high standards of software quality

Process of Test Driven Development (TDD)



- Run all the test cases and make sure that the new test case fails.
- Red – Create a test case and make it fail, Run the test cases
- Green – Make the test case pass by any means.
- Refactor – Change the code to remove duplicate/redundancy.Refactor code – This is done to remove duplication of code.
- Repeat the above-mentioned steps again and again

Advantages of Test Driven Development

- Unit test provides constant feedback about the functions.
- Quality of design increases which further helps in proper maintenance.
- Test driven development act as a safety net against the bugs.
- TDD ensures that your application actually meets requirements defined for it.
- TDD have very short development lifecycle.

Disadvantages of TDD

- Increased Code Volume: Using TDD means writing extra code for tests cases , which can make the overall codebase larger and more Unstructured.
- False Security from Tests: Passing tests will make the developers think the code is safer only for assuming purpose.
- Maintenance Overheads: Keeping a lot of tests up-to-date can be difficult to maintain the information and its also time-consuming process.
- Time-Consuming Test Processes: Writing and maintaining the tests can take a long time.
- Testing Environment Set-Up: TDD needs to be a proper testing environment in which it will make effort to set up and maintain the codes and data.

Pair Programming

- Pair programming is a development technique in which two programmers work together at a single workstation.
- A person who writes code is called a driver and a person who observes and navigates each line of the code is called a navigator.
- They may switch their role frequently. Sometimes pair programming is also known as pairing.

Pairing Variations

- Newbie-newbie pairing can sometimes give a great result because it is better than one solo newbie. But generally, this pair is rarely practiced.
- Expert–newbie pairing gives significant results. In this pairing, a newbie can learn many things from an expert, and an expert gets a chance to share his knowledge with a newbie.
- Expert–expert pairing is a good choice for higher productivity as both would be experts, so they can work very efficiently.

How does pair programming work?

- Pair programming is a collaborative software development technique where two programmers work together at one workstation.
- One assumes the role of the driver, actively typing code, while the other acts as the navigator, reviewing each line for strategic direction, potential issues, and improvements.
- They frequently switch roles to maintain engagement and share knowledge effectively.
- This method enhances code quality through immediate error detection and fosters communication, speeding up problem-solving and reducing knowledge silos within the team.

Refactoring

- A refactoring is a “behavior-preserving transformation” – Joshua Kerievsky
- Refactoring is “a change made to the internal structure of software to make it easier to understand and cheaper to modify without changing its observable behavior” – Martin Fowler

- Refactoring is the practice of continuously improving the design of existing code, without changing the fundamental behavior.
- In Agile, teams maintain and enhance their code on an incremental basis from Sprint to Sprint.
- If code is not refactored in an Agile project, it will result in poor code quality, such as unhealthy dependencies between classes or packages, improper class responsibility allocation, too many responsibilities per method or class, duplicate code, and a variety of other types of confusion and clutter.
- Refactoring helps to remove this chaos and simplifies the unclear and complex code.

1.Enlist and explain characteristics of software which describe the nature of software.
2.Explain the layered approach of software engineering
3.What are the Seven engineering principles ?Explain each in detail
4.What is software process? Explain process framework.
5.What are various umbrella activities associated in development process.
6.What do you mean by software myth
explain a.Managment b.Customers d. Practitioner Myths
7.Expalin with advantages and disadvantage of
a.Waterfall b. Incremental c.RAD d.Prototype d.Spiral Model
8.Compare the agile with evolutionary process models .
9.Explain Agile development, what are the key factors that are considered for agile development.
10.List the principle of agility
11.Explain extreme programming process.
12.Explain Scrum Process Flow.
13.Explain features of
a.JIRA Tool
b.Kanban Tool

Thank You